

WEIGHT REDUCTION AND OPTIMIZATION OF MARS ROVER PROTOTYPE

Honours Mini Project Report

Submitted in partial fulfillment of
the requirements for the
Degree of Bachelor of Technology

In

Mechanical Engineering

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CERTIFICATE

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ACKNOWLEDGEMENT

We would like to express our sincere gratitude to **Dr. Sachin Mastud**, our faculty advisor, for his unwavering support, valuable insights, and constant encouragement throughout the duration of this project. His expertise and mentorship played a key role in shaping our technical understanding and ensuring that we remained aligned with our objectives. His constructive feedback at critical stages helped us refine both our methodology and overall approach.

CERTIFICATE OF APPROVAL

This is to certify that the thesis entitled “**Weight Reduction and Optimization Of Mars Rover Prototype**” submitted by the following students is found to be satisfactory and is approved for the award of the degree of Bachelor of Technology in Mechanical Engineering from Veermata Jijabai Technological Institute, Mumbai.

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ABSTRACT

In the evolving domain of competitive robotics, structural efficiency and material optimization are critical success factors for enhancing performance while reducing operational costs. This project focuses on the **design and weight optimization** of a rover intended for intercollegiate competitions such as the **International Rover Challenge (IRC)** and the **European Rover Challenge (ERC)**. The primary objective was to develop a **cost-effective, lightweight, and structurally robust rover chassis** capable of performing key surface tasks including terrain maneuverability, object manipulation, and chemical soil analysis.

The project deployed a **dual-pronged optimization strategy: topology optimization and material optimization**. Topology optimization was employed using Finite Element Analysis (FEA) in ANSYS to strategically eliminate low-stress material zones while retaining strength-critical regions, thus achieving a **30% reduction** in chassis weight without compromising structural integrity. In parallel, material optimization was performed by substituting conventional materials such as steel with lightweight alternatives like **aluminum alloys** and **carbon fiber composites**, significantly improving the strength-to-weight ratio.

Key structural components targeted included the **chassis, robotic arm, and suspension system** (notably the **differential and rocker-bogie linkages**). The optimized design outcomes demonstrated enhanced **mechanical stiffness**, improved **traction performance**, and reduced **manual transport effort**, achieving a **total weight reduction of 33%** across the rover's main load-bearing structures.

Advanced validation was performed using the Tsai-Wu failure theory to ensure composite material safety under operational loads. The final optimized model, combining high structural resilience with minimized mass, provides a robust engineering framework suitable for real-world deployment and competitive edge in rover design challenges. This project exemplifies the effective synergy of engineering innovation, computational modeling, and material science to drive performance improvements and operational efficiencies in modern rover systems.

[**keywords** : Rover, Robotic Arm, Suspension, Differential, Chassis]

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1.2 REPORT OUTLINE

Chapter 1: Introduction

This chapter discusses the Mars Rover's role in space exploration and its importance in scientific research, particularly on planetary surfaces. It highlights the use of student prototypes in advancing rover technology and their contribution to broader aerospace engineering.

Chapter 2: Literature Review

A review of existing rover designs with a focus on chassis, differential, robotic arm, and suspension systems. It explores various chassis types, differential gear mechanisms, and robotic arm designs used in planetary exploration.

Chapter 3: Model Development (CAD)

Covers the CAD development of the chassis, differential, robotic arm, and suspension system. It also includes material selection for each component and discusses how these materials contribute to overall weight optimization and structural integrity.

Chapter 5: Simulation

This chapter discusses the simulation results for the chassis, differential, robotic arm, and suspension system. It includes meshing, boundary conditions, and deformation analysis for both optimized and unoptimized designs.

Chapter 6: Results and Conclusion

Summarizes the optimized results, focusing on the weight reduction, performance improvements, and structural efficiency

CHAPTER 1: INTRODUCTION

1.1 MARS ROVER: FUNCTION IN SPACE EXPLORATION

The quest for planetary exploration has consistently driven advancements in robotics and engineering design. Mars rovers, in particular, serve as critical agents for the scientific exploration of extraterrestrial surfaces. These autonomous or semi-autonomous platforms are tasked with traversing rugged terrain, conducting soil and atmospheric analyses, capturing high-resolution imagery, and enabling remote experimentation in hostile environments where direct human presence is unfeasible.

Rovers such as *Sojourner*, *Spirit*, *Opportunity*, *Curiosity*, and *Perseverance* have contributed immensely to our understanding of Mars' geological history, potential for past life, and resource availability for future manned missions. They embody a convergence of multidisciplinary engineering — mechanical systems for mobility, electrical systems for power distribution, and computational systems for autonomous navigation — operating seamlessly in highly constrained and unpredictable settings. Consequently, rover design encapsulates a gold standard of engineering excellence, integrating structural optimization, thermal management, material science, and mechatronics under extreme conditions.

1.2 IMPORTANCE OF STUDENT PROTOTYPES FOR RESEARCH ADVANCEMENT

Student-driven rover prototypes, often designed for competitions such as the **International Rover Challenge (IRC)** and **European Rover Challenge (ERC)**, play an essential role in advancing research and development in space robotics. These initiatives serve as incubators for innovation, offering a controlled yet challenging platform for testing novel ideas, structural optimization techniques, material applications, and advanced simulation methodologies like Finite Element Analysis (FEA) and Topology Optimization.

Beyond their educational value, student-built prototypes contribute meaningfully to the broader aerospace research ecosystem. They enable low-cost experimental validation of theoretical models, foster interdisciplinary collaboration, and often serve as early-stage feasibility studies for design concepts that could later be adapted for real space missions. The iterative process of designing, fabricating, and testing prototypes cultivates critical competencies in systems thinking, failure analysis, and design for manufacturability — essential skill sets for next-generation aerospace engineers.

Moreover, the constrained budgets and aggressive timelines inherent to student projects drive a culture of efficiency, rapid innovation, and practical problem-solving, mirroring the resource-limited conditions faced in actual space missions. Thus, student rovers are not merely academic exercises but significant stepping stones for future space exploration technologies.

1.3 CONTRIBUTION TO BROADER AERONAUTICAL AND ASTRONAUTICAL ENGINEERING FIELDS

The challenges associated with designing effective rover systems transcend the boundaries of mechanical engineering and deeply impact other facets of **aeronautical and astronautical engineering**. Lessons learned from rover development inform critical aspects such as:

- **Lightweight structural design**, directly influencing aircraft and spacecraft development.
- **Thermal management systems**, applicable in spacecraft subjected to extreme temperature fluctuations.
- **Mobility and control systems**, feeding into autonomous aircraft (UAV) and planetary lander designs.
- **Reliability engineering and redundancy modeling**, critical for both manned and unmanned aerospace missions.

Additionally, the adoption of **advanced materials** like carbon composites and high-strength aluminum alloys, along with computational methods like **multiphysics simulations**, directly reflects the trajectory of modern aerospace innovation. Student rover projects thus act as a vital pipeline, ensuring that novel concepts in aeronautical engineering are not confined to theoretical studies but are rigorously validated through hands-on experimentation and deployment in complex operational environments.

In this context, the present project—focused on **structural optimization and material enhancement of a Mars rover prototype**—contributes to both the immediate objectives of rover design and the broader, forward-looking ambitions of the aeronautical and astronautical engineering industries.

CHAPTER 2: LITERATURE REVIEW

2.1 CHASSIS

2.1.1 INTRODUCTION

The chassis is the central structural framework of a planetary rover, providing the foundation upon which all critical subsystems, including the suspension system, robotic arm, power modules, scientific payloads, and communication units, are mounted. Beyond serving as a mounting platform, the chassis plays a vital role in maintaining the mechanical integrity and operational stability of the rover as it navigates diverse and unpredictable terrains. In planetary exploration missions, where environmental conditions are harsh and maintenance is impossible, the chassis must be designed to withstand static, dynamic, and thermal loads while contributing minimally to the overall mass of the vehicle. An effective chassis design not only ensures structural robustness but also enables the rover to fulfill its mission objectives efficiently and reliably across extended operational periods.

2.1.2 FUNCTIONS

The rover chassis serves multiple critical functions that are essential to mission success:

- **Structural Support:**

The chassis supports and connects all major subsystems, ensuring they remain properly aligned and secured during dynamic operations.

- **Load Distribution:**

It evenly distributes static arising from terrain interactions, robotic operations, and payload handling, preventing localized stress concentrations that could cause structural failures.

- **Protection:**

The chassis acts as a shield, protecting internal electronics and scientific equipment from mechanical shocks, vibrations, dust infiltration, and thermal fluctuations experienced during planetary exploration.

2.1.2 TYPES

Different mission requirements, terrain expectations, and payload needs have led to the development of various chassis designs for planetary rovers. The three main types include:

1. Monocoque Chassis

- A monocoque chassis is an integrated structural design where the external skin itself bears the mechanical loads. This approach minimizes mass while maximizing stiffness, as the body and frame are merged into a single unified structure. Monocoque designs are typically used in smaller rovers or those where weight savings are critical, such as lunar micro-rovers or compact Mars scouts. Although offering excellent weight efficiency, monocoque structures can be challenging to repair and are less modular compared to frame-based designs.

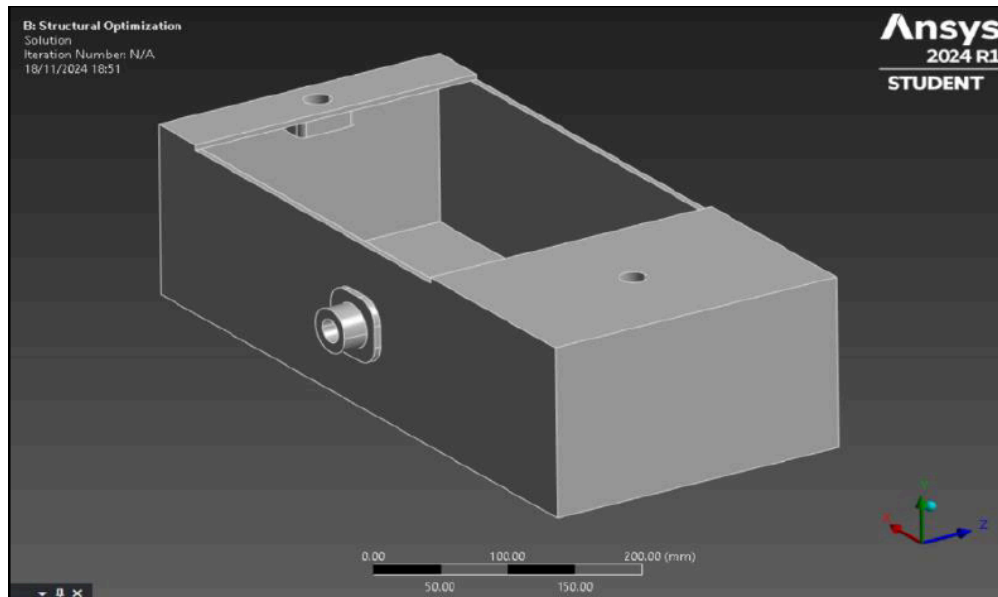


Figure 2.1: Sheet Welded Monocoque Chassis.

2. Space Frame Chassis

- Space frame chassis are constructed from a network of tubular members. This design distributes loads efficiently throughout the structure, offering high torsional stiffness and resistance to dynamic deformations. Space frames are particularly well-suited for medium to large rovers that need to endure significant terrain impacts, robotic arm operations, or heavier payloads. Additionally, the modular nature of space frames allows for easier repairs and subsystem replacements during rover development phases.

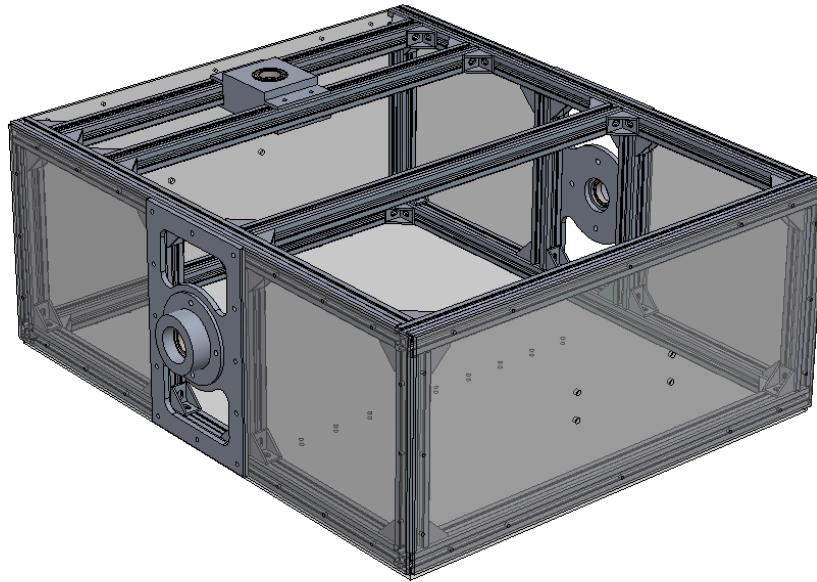


Figure 2.2: Space Frame Chassis.

2.2 DIFFERENTIAL

2.2.1 INTRODUCTION

In planetary exploration rovers, particularly those engineered for traversing extreme and unpredictable terrains such as those encountered on Mars, the differential is a vital mechanical component within the suspension system architecture. The differential serves as a passive but crucial element ensuring the stability of the chassis during complex terrain navigation. It provides a mechanical linkage between the left and right suspension arms, enabling coordinated vertical movement that maintains the rover's body in a relatively level orientation even when encountering severe surface irregularities. This passive mechanical stability is critical for preserving the functional integrity of sensitive scientific instruments mounted on the rover, preventing damage due to excessive tilting, and ensuring consistent mobility across rugged landscapes. Unlike systems that require active control through electronics or hydraulics, the rover differential operates purely mechanically, offering high reliability and fail-safe performance with minimal maintenance needs. Its design simplicity, low energy dependency, and robust behavior under dynamic environmental stresses make it an indispensable part of the rover's suspension strategy, directly contributing to overall mission success by improving maneuverability, safety, and operational durability over extended mission lifetimes.

2.2.2 FUNCTIONAL OVERVIEW

The rover differential acts as a mechanical stabilizer linking the left and right rocker arms, allowing relative vertical movement without introducing rotational forces or requiring active control inputs. Mechanically, the differential permits the angular displacement between suspension arms: when one side of the rover climbs over an obstacle causing its rocker arm to rise, the differential mechanism simultaneously accommodates a downward movement on the opposite side. This mechanical balance ensures that the center of mass of the rover remains approximately stable relative to the ground, significantly reducing unwanted chassis tilt and rotational instability. The differential's operation enhances all-wheel contact with the ground surface, maximizing traction and minimizing wheel slip — critical parameters for mission endurance on loose, rocky, or sandy extraterrestrial soils. Furthermore, by evenly distributing loads across the rover structure, it prevents localized stress concentrations, thereby prolonging the mechanical health of suspension links, joints, and frames. The differential does not transmit driving torque; it neither powers the wheels nor controls their speeds. Its sole objective is to passively harmonize suspension dynamics during terrain negotiation, ensuring smooth operation with no reliance on electronic control systems or actuators. In essence, the differential forms the backbone of the rover's dynamic stability system, silently enabling the vehicle to carry out complex navigation tasks with minimal human intervention and maximum mechanical resilience, making it a cornerstone of modern planetary rover architecture.

2.2.3 TYPES

1. Passive mechanical differential

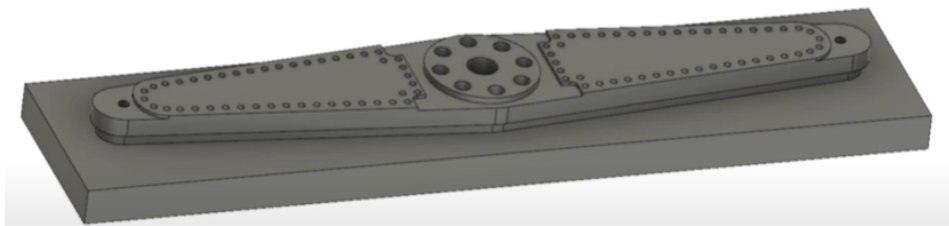


Figure 2.3: Passive differential

- This is the most commonly used differential in planetary rovers. It utilizes a mechanical linkage that passively balances the rover's body as it traverses uneven terrain. When one side of the suspension rises over an obstacle, the differential causes the opposite side to lower, maintaining the chassis's level orientation.

2. Differential gearbox

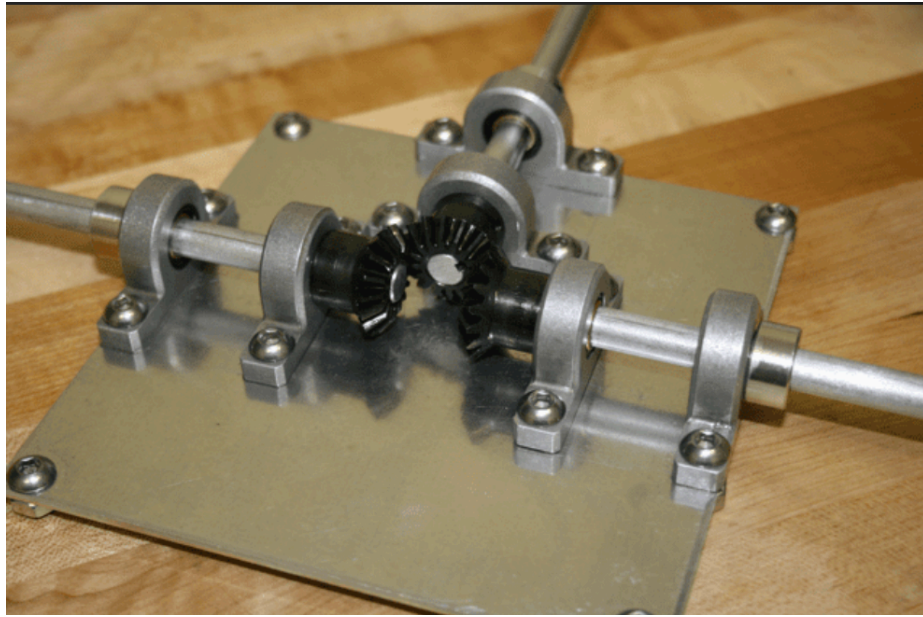


Figure 2.4 : Differential gearbox.

- Used in earlier Mars rovers like Spirit and Opportunity, the differential gearbox connects the rocker arms through a set of gears. This setup allows for smoother distribution of motion between the suspension arms, enhancing stability over rugged terrain.^[2]

2.3 ROBOTIC ARM^[3]

2.3.1 INTRODUCTION

A robotic arm is a type of mechanical arm, usually programmable, with similar functions to a human arm; the arm may be the sum total of the mechanism or may be part of a more complex robot. The links of such a manipulator are connected by joints allowing either rotational motion (such as in an articulated robot) or translational (linear) displacement. The links of the manipulator can be considered to form a kinematic chain. The terminus of the kinematic chain of the manipulator is called the end effector and it is analogous to the human hand. However, the term "robotic hand" as a synonym of the robotic arm is often proscribed.

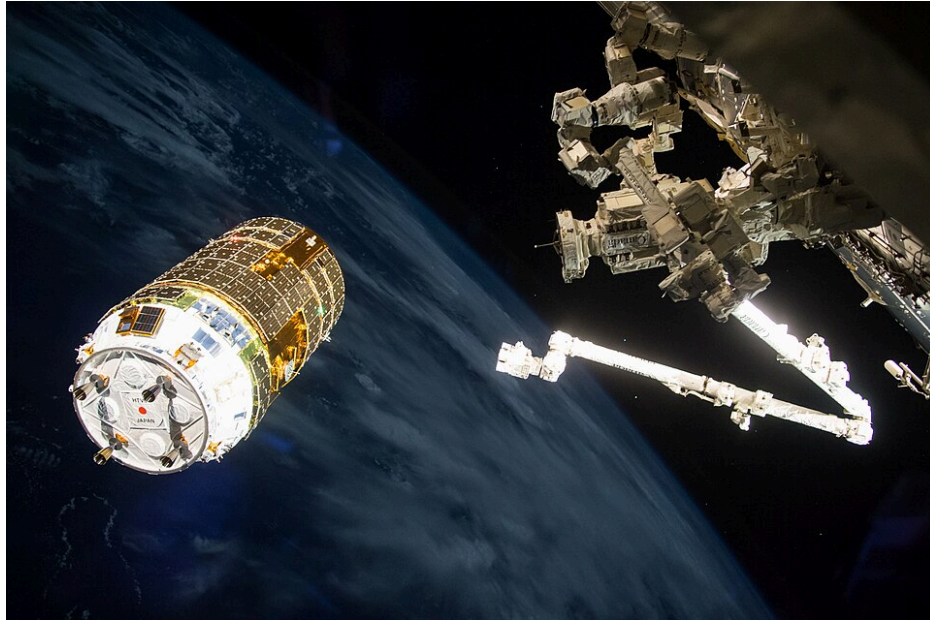


Figure 2.5 : Canadarm

In space, the Canadarm and its successor Canadarm2 are examples of multi degree of freedom robotic arms. These robotic arms have been used to perform a variety of tasks such as inspection of the Space Shuttle using a specially deployed boom with cameras and sensors attached at the end effector, and also satellite deployment and retrieval manoeuvres from the cargo bay of the Space Shuttle. The Curiosity and Perseverance rovers on the planet Mars also use robotic arms. Additionally, Perseverance has a smaller sample caching arm hidden inside its body below the rover in its caching assembly. TAGSAM is a robotic arm for collecting a sample from a small asteroid in space on the spacecraft OSIRIS-REx.

The 2018 Mars lander InSight has a robotic arm called the IDA, it has a camera, grappler, and is used to move special instruments.

2.3.2 FUNCTIONAL OVERVIEW

In manufacturing, the robotic arm is commonly employed. Seven metal pieces linked by six joints make up a standard mechanical arm. Individual stepper motors linked to each joint are rotated by the robot's control computer (some larger arms use hydraulics or pneumatics). Step motors move in precise increments, unlike regular motors. By repeatedly executing the identical motion, the computer can manipulate the arm very accurately. The robot uses motion sensors to ensure that it moves just enough.

An industrial collaborative robot with six joints, which includes the equivalent of a shoulder, elbow, and wrist, closely mimics a human arm. Instead of being mounted on a moving body, the shoulder is often attached to a fixed base structure. With six degrees of freedom, this kind

of robot can pivot in six distinct directions. In contrast, a human arm has seven degrees of freedom.

The human arm's function is to move the hand from one location to another. Similar to this, the function of the robotic arm is to move an end effector. Robotic arms may be fitted with a variety of end effectors, each of which is appropriate for a certain purpose. A simplified hand that can grip and carry various things is one frequent end effector.

Some models of robotic hands frequently contain in-built pressure sensors that communicate to the computer how firmly the robot is holding an object. This prevents the robot from dropping or shattering the object it is holding.

2.3.3 COMPONENTS

- 1) **Links** - They are rigid structures, which transmit forces from one node of the manipulator to the other. Links are responsible for the structural integrity of the robotic arm and hold the components together, which transmit power and motion from one point to another.



Figure 2.6 : Links

- 2) **End Effector** - It is referred to as the 'hand' of a robotic arm, and is responsible for object retrieval and placing tasks. For space applications, it is used for sample retrieval and maintenance.



Figure 2.7 : End effector

- 3) **Drives** - They are responsible for providing the necessary force/torque based on their working principle. Drives make it possible to transmit motion and power, based on a given input. They can be hydraulic, pneumatic, electrical or hybrid. Drives are used in conjunction with gears, pulleys, belts, to amplify the power output/ speed output based on the requirement.



Figure 2.8 : Pneumatic Drive



Figure 2.9 : Hydraulic Drive



Figure 2.10 : Electric Drive

- 4) **Base** - The structure, which bears the entire load of the robotic arm is known as the base. It is responsible for giving an extra degree of freedom if required and for maintaining stability of the arm.

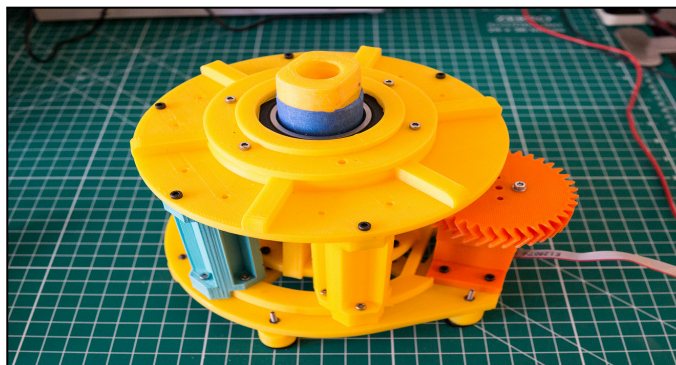


Figure 2.11 : Base

2.3.4 TYPES

- 1) **Cartesian robot / Gantry robot:** Used for pick and place work, application of sealant, assembly operations, handling machine tools and arc welding. It is a robot whose arm has three prismatic joints, whose axes are coincident with a Cartesian coordinator.



Figure 2.12 : Gantry robot

- 2) **Collaborative robot / Cobot:** Cobot applications contrast with traditional industrial robot applications in which robots are isolated from human contact. Cobot has a large variety of applications such as: commercial application, robotic research, dispensing, material handling, assembly, finishing, and quality inspection. Cobot safety may rely on lightweight construction materials, rounded edges, and the inherent limitation of speed and force, or on sensors and software that ensures safe behavior.
- 3) **Cylindrical robot:** Used for assembly operations, handling machine tools, spot welding, and handling at die casting machines. It is a robot whose axes form a cylindrical coordinate system.
- 4) **Spherical robot / Polar robot:** Used for handling machine tools, spot welding, die casting, fettling machines, gas welding and arc welding. It is a robot whose axes form a polar coordinate system.
- 5) **SCARA robot:** Used for pick and place work, application of sealant, assembly operations and handling machine tools. This robot features two parallel rotary joints to provide compliance in a plane.



Figure 2.13 : SCARA (5- axis robot)

- 6) **Articulated robot:** Used for assembly operations, die casting, fettling machines, gas welding, arc welding and spray-painting. It is a robot whose arm has at least three rotary joints.



Figure 2.14 : Articulated robot

- 7) **Parallel robot:** One use is a mobile platform handling cockpit flight simulators. It is a robot whose arms have concurrent prismatic or rotary joints.
- 8) **Anthropomorphic robot:** It is shaped in a way that resembles a human hand, i.e. with independent fingers and thumbs.

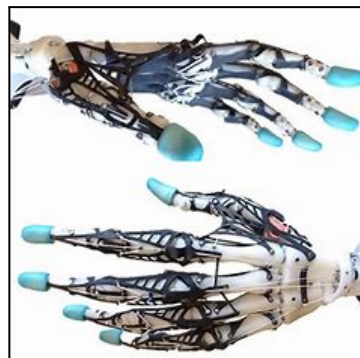


Figure 2.15 : Anthropomorphic robot

2.3.5 MECHANICAL ANALYSIS

Forward and Inverse Kinematics -

In robot kinematics, forward kinematics refers to the use of the kinematic equations of a robot to compute the position of the end-effector from specified values for the joint parameters.

The kinematics equations of the robot are used in robotics, computer games, and animation. The reverse process, that computes the joint parameters that achieve a specified position of the end-effector, is known as inverse kinematics.

The kinematics equations for the series chain of a robot are obtained using a rigid transformation $[Z]$ to characterize the relative movement allowed at each joint and separate rigid transformation $[X]$ to define the dimensions of each link. The result is a sequence of rigid transformations alternating joint and link transformations from the base of the chain to its end link, which is equated to the specified position for the end link,

$$[T] = [Z_1][X_1][Z_2][X_2] \dots [X_{n-1}][Z_n]$$

where $[T]$ is the transformation locating the end-link. These equations are called the kinematics equations of the serial chain.

A series of link transformations is applied using Denavit Hartenberg (DH) parameters technique to solve for the final solution.

$$\text{Trans}_{Z_i}(d_i) = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & d_i \\ 0 & 0 & 0 & 1 \end{bmatrix}, \quad \text{Rot}_{Z_i}(\theta_i) = \begin{bmatrix} \cos \theta_i & -\sin \theta_i & 0 & 0 \\ \sin \theta_i & \cos \theta_i & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$\text{Trans}_{X_i}(a_{i,i+1}) = \begin{bmatrix} 1 & 0 & 0 & a_{i,i+1} \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}, \quad \text{Rot}_{X_i}(\alpha_{i,i+1}) = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & \cos \alpha_{i,i+1} & -\sin \alpha_{i,i+1} & 0 \\ 0 & \sin \alpha_{i,i+1} & \cos \alpha_{i,i+1} & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$${}^{i-1}T_i = \begin{bmatrix} \cos \theta_i & -\sin \theta_i \cos \alpha_{i,i+1} & \sin \theta_i \sin \alpha_{i,i+1} & a_{i,i+1} \cos \theta_i \\ \sin \theta_i & \cos \theta_i \cos \alpha_{i,i+1} & -\cos \theta_i \sin \alpha_{i,i+1} & a_{i,i+1} \sin \theta_i \\ 0 & \sin \alpha_{i,i+1} & \cos \alpha_{i,i+1} & d_i \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

The forward kinematic equations can be used as a method in 3D computer graphics for animating models.

The essential concept of forward kinematic animation is that the positions of particular parts of the model at a specified time are calculated from the position and orientation of the object, together with any information on the joints of an articulated model. So for example if the object to be animated is an arm with the shoulder remaining at a fixed location, the location of the tip of the thumb would be calculated from the angles of the shoulder, elbow, wrist, thumb and knuckle joints. Three of these joints (the shoulder, wrist and the base of the thumb) have more than one degree of freedom, all of which must be taken into account. If the model were an entire human figure, then the location of the shoulder would also have to be calculated from other properties of the model.

Forward kinematic animation can be distinguished from inverse kinematic animation by this means of calculation - in inverse kinematics the orientation of articulated parts is calculated from the desired position of certain points on the model. It is also distinguished from other animation systems by the fact that the motion of the model is defined directly by the animator -no account is taken of any physical laws that might be in effect on the model, such as gravity or collision with other models. The present study focuses on the mechanical aspects of the robotic arm and mainly on its structural analysis and optimization.

CHAPTER 3: MODEL DEVELOPMENT (CAD)

3.1 CHASSIS

3.1.1 CAD FOR CHASSIS

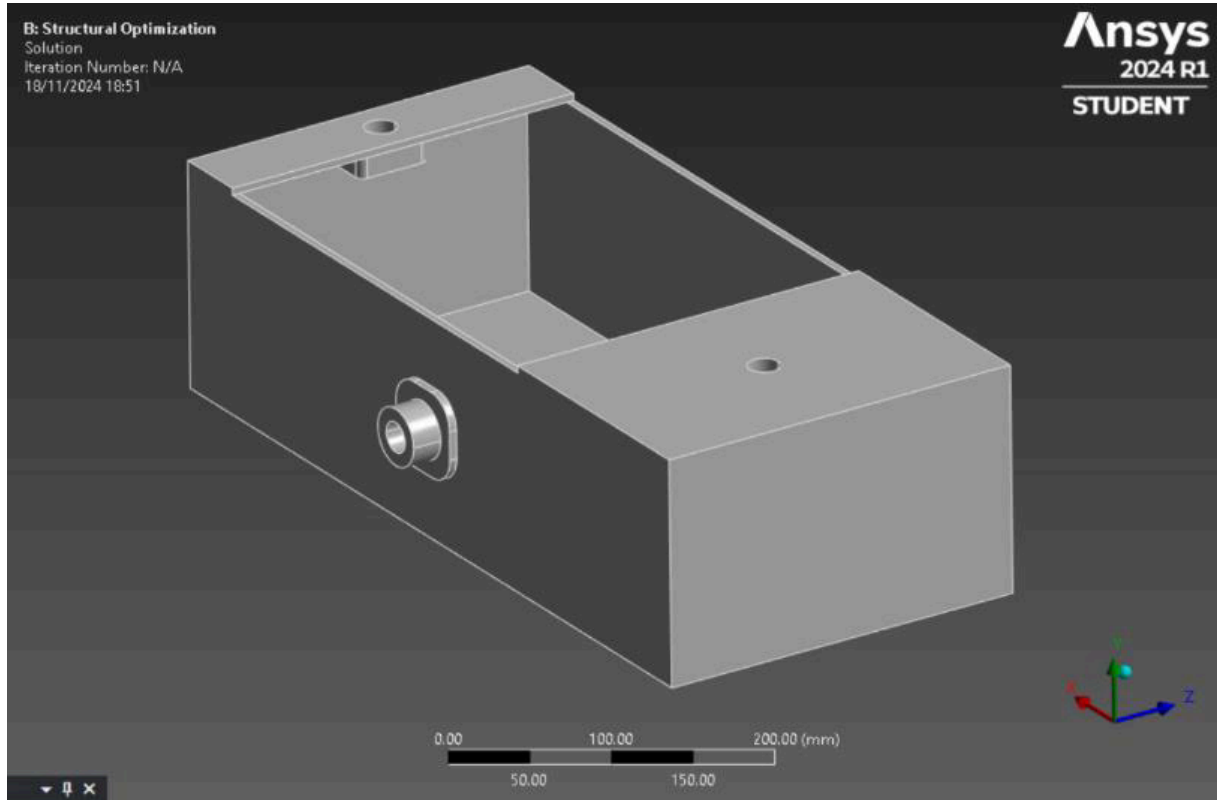


Figure 3.1: Design for unoptimised chassis.

The provided geometry represents the chassis of a rover, specifically designed to serve as the primary structural foundation for housing and supporting key subsystems. With dimensions of 540 mm in length, 250 mm in breadth, and 160 mm in height, the chassis is compact yet robust, ensuring a balance between lightweight design and structural durability for mobility and off-road performance.

A cylindrical structure on one side of the chassis houses the bearings that support the suspension shaft, playing a critical role in accommodating dynamic forces and enabling smooth suspension operation. This design ensures that the suspension system is securely integrated into the chassis, providing stability and flexibility for handling uneven terrain.

The top base at the front is dedicated to mounting the arm, which is a mechanical component used for tasks such as picking or manipulating objects. This mounting area ensures that the arm is securely fixed and aligned for precise operations. On the other hand, the top base at the back is designed to mount the differential, a vital component for keeping wheels in contact with the surface. Enabling the rover to navigate challenging environments.

It is made using sheet metal which is lazer-cut and welded together.

3.1.2 MATERIAL SELECTION FOR CHASSIS

Why is aluminum selected in comparison with steel for the chassis?

Aluminium is chosen for the rover chassis over structural steel due to its significant weight advantage. With a density of approximately 2.7 g/cm^3 compared to steel's $\sim 7.8 \text{ g/cm}^3$, aluminium offers a lighter structure, which improves the rover's mobility and energy efficiency. Additionally, aluminium is naturally corrosion-resistant, eliminating the need for protective coatings required for steel in harsh environments. While steel is stronger in absolute terms, aluminium's superior strength-to-weight ratio makes it a practical choice for lightweight applications. Moreover, aluminium is easier to machine, cut, and weld, simplifying the fabrication process compared to steel, which demands more robust tools and techniques.

Why is aluminium selected in comparison to Carbon Fibre for the chassis?

Compared to carbon fibre, aluminium is far more cost-effective, making it suitable for budget-conscious projects like the rover. While carbon fibre is lighter and stronger in certain directions, it is brittle and prone to cracking under impact, whereas aluminium can plastically deform and absorb energy without catastrophic failure. Aluminium also offers excellent thermal conductivity, beneficial for dissipating heat from electronics, an advantage carbon fibre, being an insulator, lacks. Furthermore, aluminium structures are easier to repair through welding or reinforcement, while carbon fibre requires complex fabrication or replacement.

Properties of Outline Row 4: Aluminum Alloy			
	A	B	C
1	Property	Value	Unit
2	Density	2770	kg m ⁻³
3	Isotropic Secant Coefficient of Thermal Expansion		
4	Coefficient of Thermal Expansion	2.3E-05	C ⁻¹
5	Isotropic Elasticity		
6	Derive from	Young's Modulus and Poi...	
7	Young's Modulus	7.1E+10	Pa
8	Poisson's Ratio	0.33	
9	Bulk Modulus	6.9608E+10	Pa
10	Shear Modulus	2.6692E+10	Pa
11	S-N Curve	Tabular	
15	Tensile Yield Strength	2.8E+08	Pa
16	Compressive Yield Strength	2.8E+08	Pa
17	Tensile Ultimate Strength	3.1E+08	Pa

Figure 3.2: Properties of Aluminium.

3.2 DIFFERENTIAL

3.2.1 CAD FOR DIFFERENTIAL

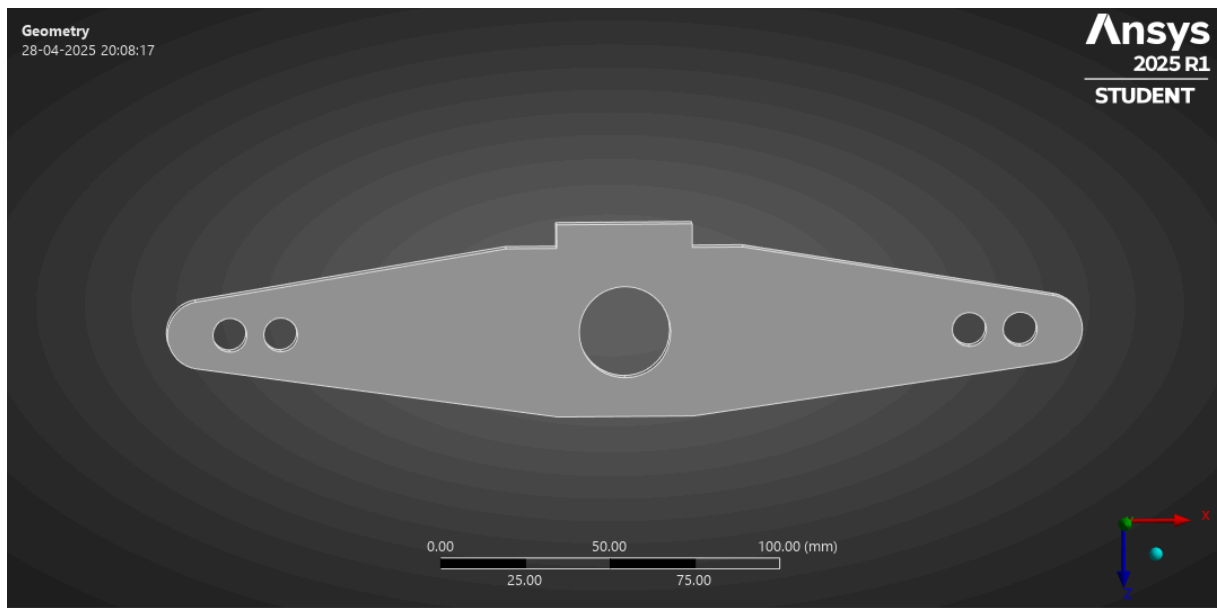


Figure 3.3 : Unoptimized Differential

The starting point for the differential design was to create a simple, strong, and easily manufacturable structure that could effectively balance the movement of the rover's suspension arms. The differential was modeled as a straight rectangular bar, measuring 240 mm in length, sized to fit neatly across the chassis frame and align properly with the rocker-bogie suspension setup. A base plate of 120 mm was added at the center, with bearing holes of 15 mm diameter on both sides to allow smooth articulation where the suspension links connect. The geometry was kept intentionally straightforward, focusing on providing enough mechanical strength while avoiding unnecessary complexity that could lead to stress concentrations or fabrication challenges.

3.2.1 MATERIAL SELECTION FOR DIFFERENTIAL

The material selection for the rover differential was based on balancing mechanical strength, durability, and weight efficiency — all of which are critical factors in planetary rover applications. After evaluating several options, aluminum alloy Al-6061 was selected for the differential component. This material offers a favorable combination of properties that align well with the mission objectives.

Al-6061 provides an excellent strength-to-weight ratio, ensuring that the differential remains structurally stable without unnecessarily increasing the rover's overall mass. In addition,

Al-6061 is relatively easy to machine and fabricate, making it a practical choice for prototype manufacturing and final assembly within budget and time constraints.

From a mechanical standpoint, Al-6061 delivers sufficient yield strength and fatigue resistance to withstand the dynamic loads encountered during rover movement across rugged terrains. Its thermal conductivity also helps in managing any minor thermal stresses induced during daytime-night time temperature cycles on planetary surfaces. Considering these factors, Al-6061 emerged as the most suitable material for the unoptimized differential design, providing a good balance between strength, weight, reliability, and manufacturability.

3.3 ROBOTIC ARM

3.3.1 CAD MODEL FOR ROBOTIC ARM

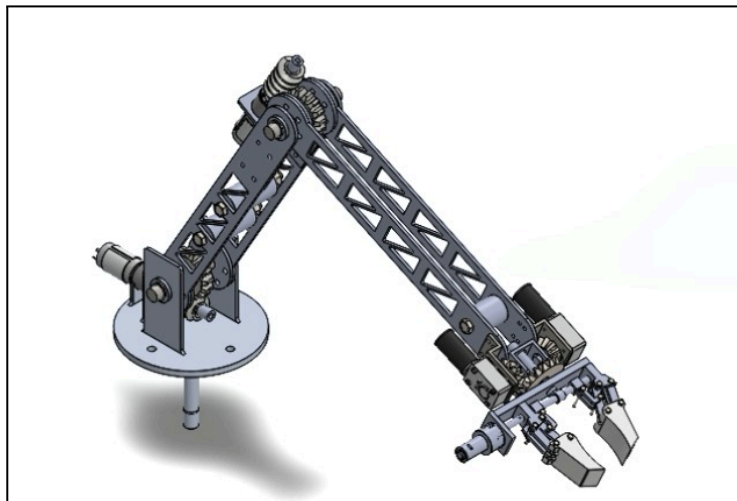


Figure 3.4 : Assembly of robotic arm

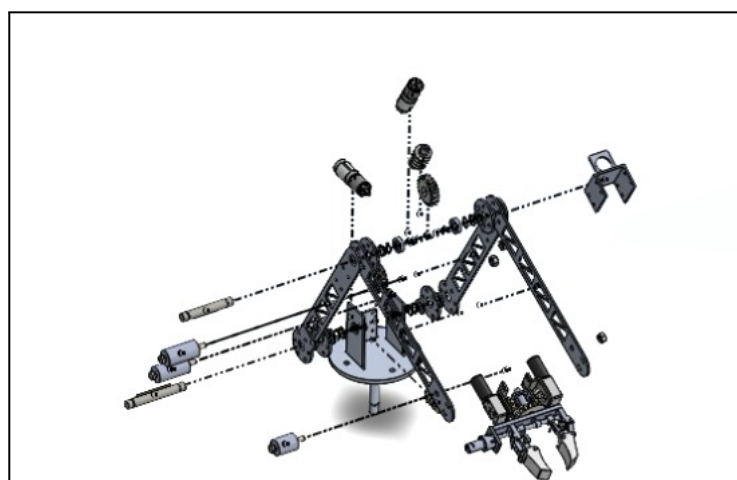


Figure 3.5 : Exploded View

3.3.2 OPTIMIZABLE PARTS

- 1) Link - The robotic arm is composed of 4 rigid links of length 400 mm each and 5 mm thick. The length of the links was decided on the basis of suspension and chassis height and also considering the length of the end effector.

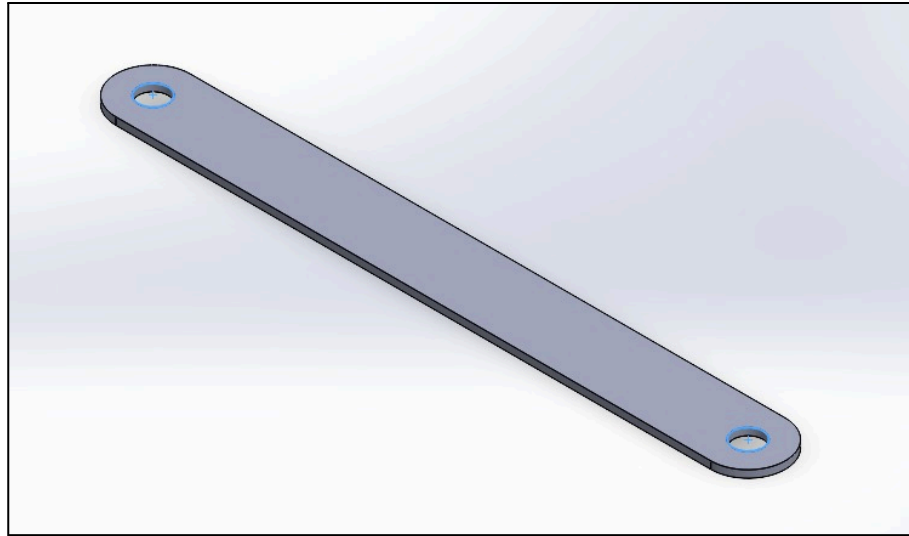


Figure 3.6 : CAD model of link

- 2) Base - The base of the arm is 200 mm in diameter and 5 mm thick. The plates on top of the base are welded, to provide firm support to the entire structure. The plates are 120 mm in length and 5 mm thickness, and consist of bearing inserts of 30 mm diameter.

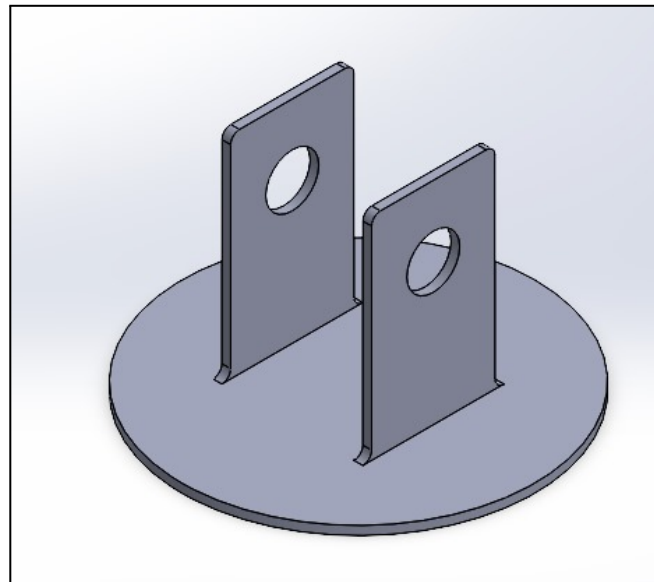


Figure 3.7 : CAD model of Base

- 3) Bevel Gear - The bevel gear system uses rolling and pitching motion of the end effector. It consists of 2 pinions and 1 gear, of the same size and module.

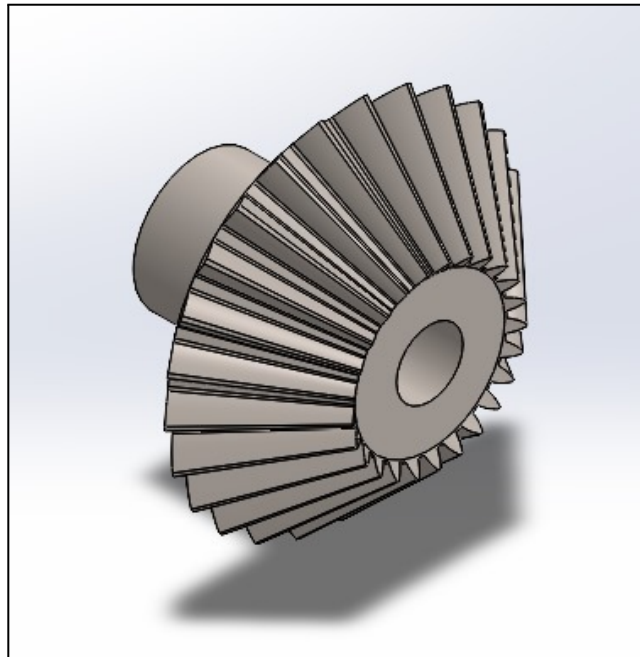


Figure 3.8 : CAD model of gear

3.3.3 MATERIAL SELECTION

Aluminium Alloy was used as the starting material for the simulation and manufacturing of the link and base. The bevel gear was manufactured using standard stainless steel.

Properties	Aluminium 6061	Stainless steel
1. Density	2.70 g/cm ³	7.90 g/cm ³
2. Young's modulus	68.9 GPa	193 GPa
3. Poisson's ratio	0.33	0.30
4. Yield strength	275 Mpa	250 Mpa
5. UTS	310 MPa	520 MPa

Table 3.1 : Material properties

3.3.4 LOAD CALCULATIONS

For subsequent analysis, the loads acting on the base, links were calculated as follows -

1) Links -

Maximum payload weight = 5 kg = 50 N

Assuming the arm stretches at a full length, considering an end effector length of 130 mm,

Therefore, total length of the arm = 930 mm

Torque = Force * Distance = 50 * 0.93 = 46.5 Nm, on the shoulder joint. Each arm would experience 23.25 Nm.

2) Base -

The base would experience the same torque due to coincident contact points between shoulder joint and bearing insert locations, as well as a load equal to the arm's weight would be acting on it, along with the payload. The final weight of the arm is found to be approximately 15 kg, which is equal to 150 N.

3) Bevel system -

The bevel pinion and gear would experience only the weight of the payload, since rolling and pitching would be restricted in such scenarios.

3.4 SUSPENSION

The differential and rocker-bogie suspension system used in the rover is a critical component. It plays a unique role in maintaining the stability and balance of the rover during traversal over uneven terrains. Its primary function is to ensure that the centre of mass remains stable by counterbalancing the effects of vertical wheel displacement. This stability is essential for manoeuvring in rugged environments. The differential facilitates the alignment of the suspension system, contributing to the rover's ability to steer effectively and maintain directional control. The "rocker" part of the suspension comes from the rocking aspect of the larger, body-mounted linkage on each side of the rover. These rockers are connected to each other and the vehicle chassis through a differential. Relative to the chassis, the rockers will rotate in opposite directions to maintain approximately equal wheel contact. The chassis maintains the average pitch angle of both rockers. One end of a rocker is fitted with a drive wheel, and the other end is pivoted to the bogie.

The "bogie" part of the suspension refers to the smaller linkage that pivots to the rocker in the middle and which has a drive wheel at each end. Bogies were commonly used as load wheels in the tracks of army tanks as idlers distributing the load over the terrain, and were also quite commonly used in trailers of semi-trailer trucks. Both tanks and semi-trailers now prefer trailing arm suspensions.

The rocker-bogie design is unsprung and uses split rather than full-width axles, allowing the rover to climb over obstacles (such as rocks) that are up to twice the wheel's diameter in size while keeping all six wheels on the ground. As with any suspension system, the tilt stability is limited by the height of the center of gravity. Systems using springs tend to tip more easily as the loaded side yields. Based on the center of mass, the Curiosity rover of the Mars Science Laboratory mission can withstand a tilt of at least 45 degrees in any direction without overturning, but automatic sensors limit the rover from exceeding 30 degree tilts. The system is designed to be used at slow speed of around 10 centimetres per second (3.9 in/s) so as to minimize dynamic shocks and consequential damage to the vehicle when surmounting sizable obstacles.

In order to go over a vertical obstacle face, the front wheels are forced against the obstacle by the center and rear wheels. The rotation of the front wheel then lifts the front of the vehicle up and over the obstacle. The middle wheel is then pressed against the obstacle by the rear wheels and pulled against the obstacle by the front until it is lifted up and over. Finally, the rear wheel is pulled over the obstacle by the front two wheels. During each wheel's traversal of the obstacle, forward progress of the vehicle is slowed or completely halted. This is not an issue for the operational speeds at which these vehicles have been operated to date.

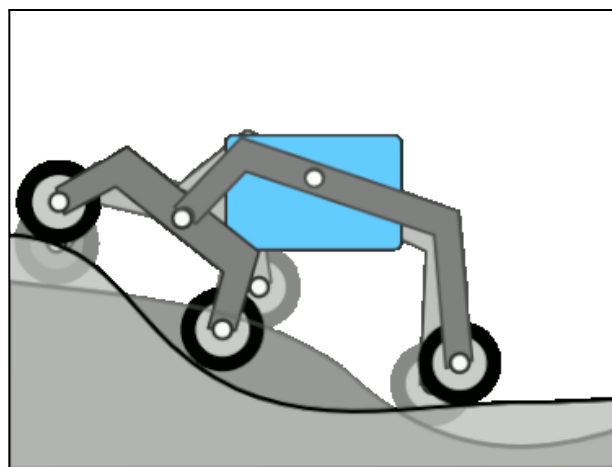


Figure 3.9 : Rocker bogie

A brief description of new mechanisms - Lambda Mechanism

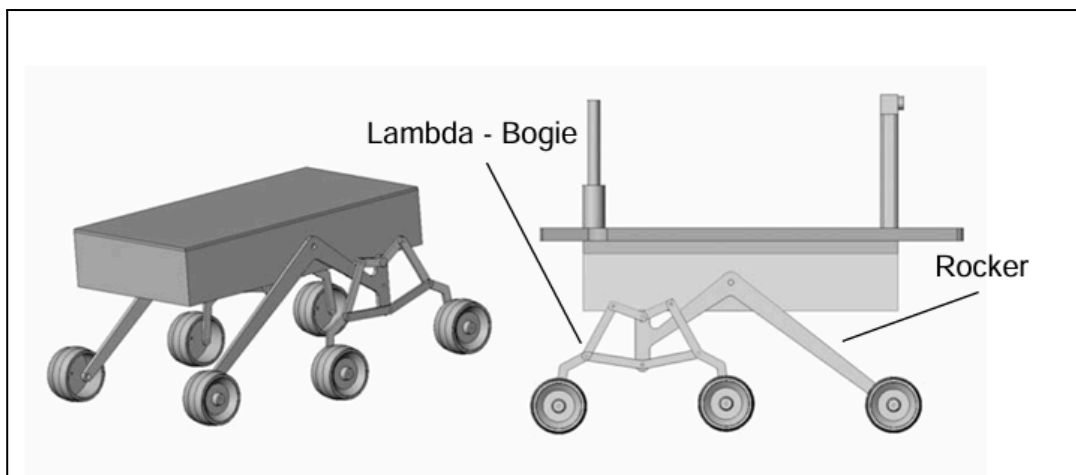
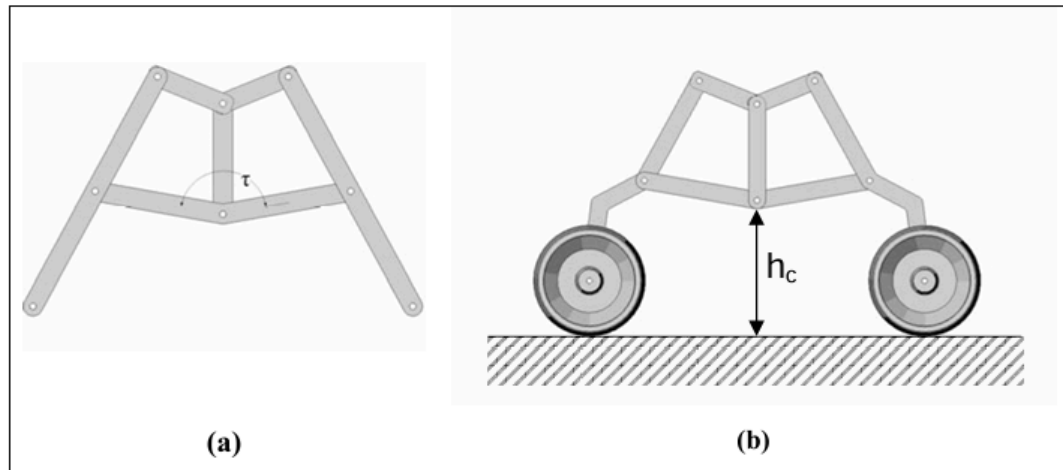


Figure 3.10 : Lambda mechanism

New bogie design consists of two lambda mechanisms which are connected symmetrically. Thus, wheels move in a straight line but in opposite directions. This design balances the reaction forces on each wheel; therefore the traction force remains the same for each wheel whether one wheel is in the upper position. Symmetric connection of two mechanisms is a critical process. Since both sides of the bogie will work in the linear part of the curve, one side will be in the opposite position of the other side.

Symmetric lambda mechanisms are connected to each other with a V-shaped rigid link. Angle τ can be selected geometrically. The constraint of this angle is ground clearance of bogie (h_c) and maximum obstacle capacity.

3.4.1 ROCKER

Rocker is the link between the differential and the wheel hubs and helps in motion transmission from the differential to the ground. It is connected to the chassis and is two-branched. One is connected to the bogie and the other to the wheel hub.

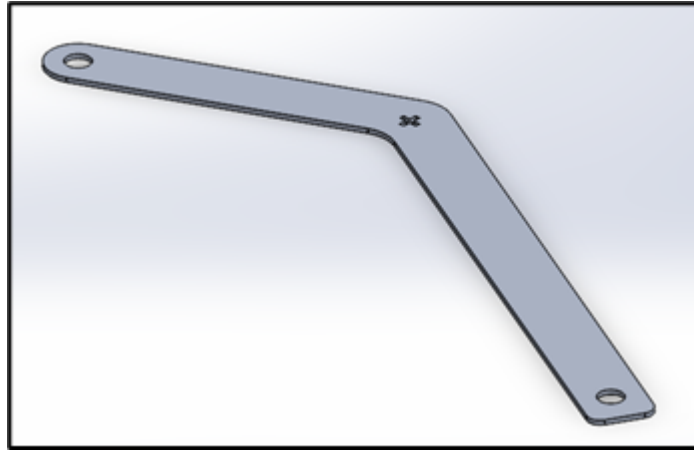


Figure 3.9 : CAD model of Rocker link

3.4.2 BOGIE

The bogie is a part of the suspension, connected to the main link using a bearing. It helps in stabilizing the path of the rover, in case of irregular terrain.

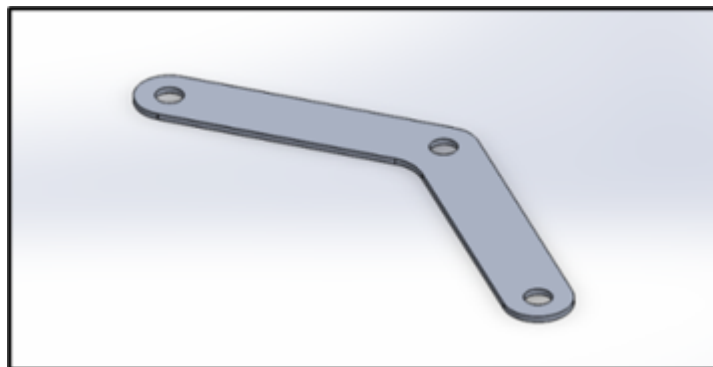


Figure 3.10 : CAD model of bogie

3.3.4 LOAD CALCULATIONS

For subsequent analysis, the loads acting on the base, links were calculated as follows -

1) Links -

Maximum payload weight = 60 kg = 600 N

The load is distributed equally between the two suspension points (300 N)

Each branch experiences half of this load, 150 N.

The angle between the branches is 45 degrees, therefore the cosine component acts on each branch, which is 106.7 N.

From the bogie's perspective, the load is distributed among 6 wheels each, thus giving a load of 100 N. The bogie experiences this at an angle of 45 degrees, which is 70.7 N.

CHAPTER 4: SIMULATION

4.1 CHASSIS

4.1.1 MESHING

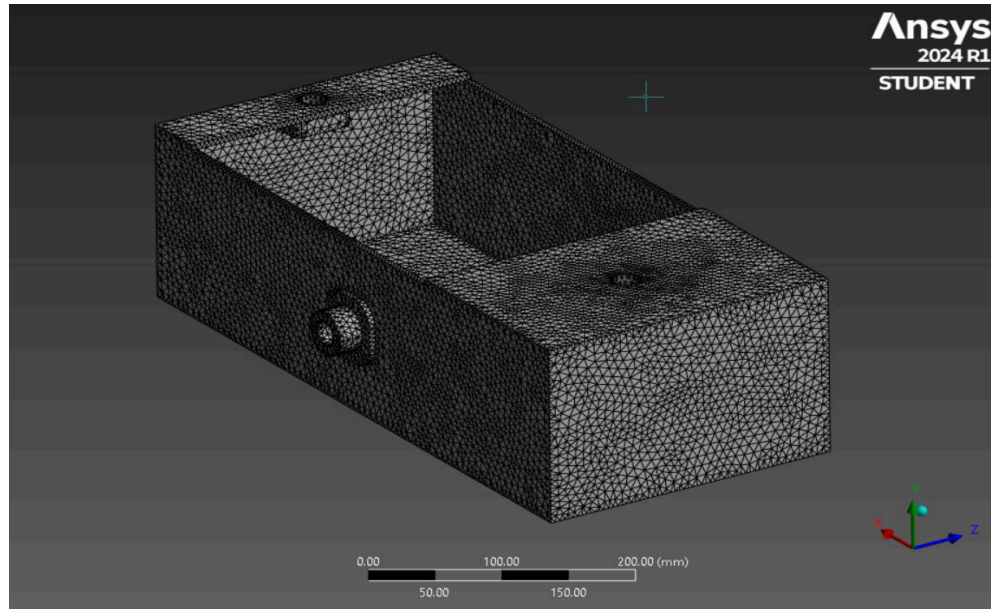


Figure 4.1: Meshing Performed on the chassis.

The mesh is generated to discretize the geometry of the model for finite element analysis (FEA). By dividing the model into smaller elements (triangular in this case), the software can numerically solve the physical equations governing the model's behavior (such as stress, strain, and deformation) under applied loads. A finer mesh, with more elements, provides more accurate results, but at the cost of increased computational resources. This mesh helps in simulating and analyzing how the structure will respond to forces, moments, and boundary conditions.

Mesh Statistics: The mesh contains 1,27,279 nodes and 64,221 elements, which are parameters that define the complexity and resolution of the mesh.

4.1.2 FORCES ACTING

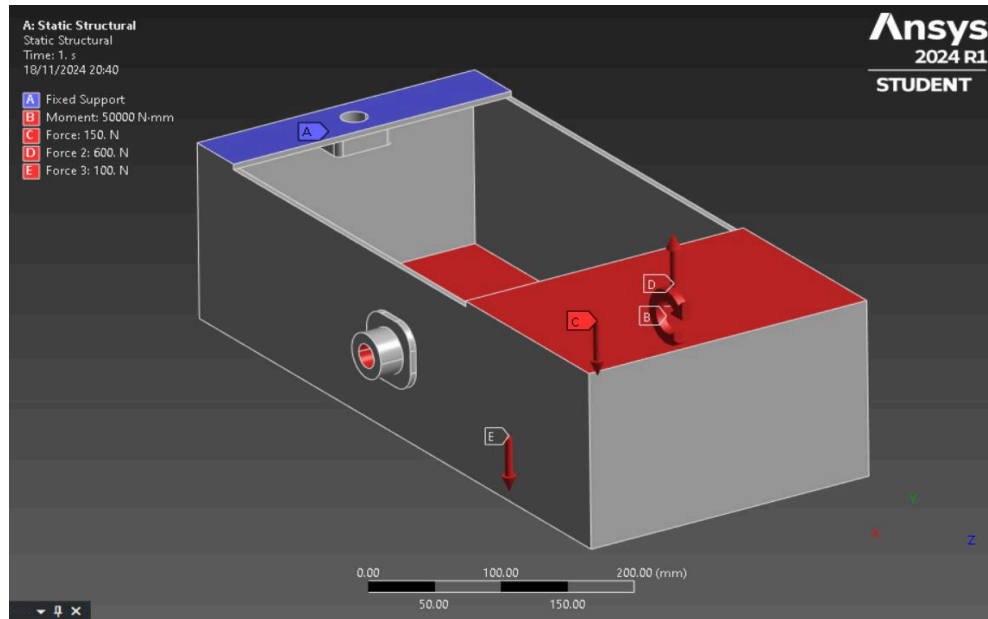


Figure 4.2: Forces acting on the chassis.

Moment (B): The moment of **50,000 N·mm** is due to the **robotic arm's fully extended state** when lifting the **5 kg maximum weight**. The arm links are **1000 mm** long, and gravity is **10 m/s²**, which generates a rotational force (torque) about the pivot point as the arm extends. The moment is calculated by multiplying the force (**5 kg * 10 m/s² = 50 N**) by the length of the arm (**1000 mm = 1 m**), so the moment is **50 N * 1 m = 50 N·m or 50,000 N·mm**.

Force 1 (C): The **self-weight** of the robotic arm, which is **15 kg**, is applied as a vertical force. The force due to gravity is calculated as **15 kg * 10 m/s² = 150 N**. This represents the downward force exerted by the arm's weight.

Force 2 (D): The **reaction force by the suspension on the chassis** is applied. This force is likely the reactive load due to the interaction between the chassis and the suspension, ensuring the structure remains stable under the applied forces.

Force 3 (E): The **weight of the electronics** placed on the **base of the chassis**, which could be an additional load acting downward on the structure, is likely included in the analysis to ensure the chassis can bear the load of all components. The weight of the electronics would be calculated similarly by multiplying its mass by gravity.

It is important to note that, For the factor of safety (FoS), consideration of multiple worst-case scenarios to ensure the structural integrity of the chassis under extreme load conditions.

4.1.3 ANALYSIS FOR UN-OPTIMISED DESIGN

4.1.3.1 TOTAL DEFORMATION

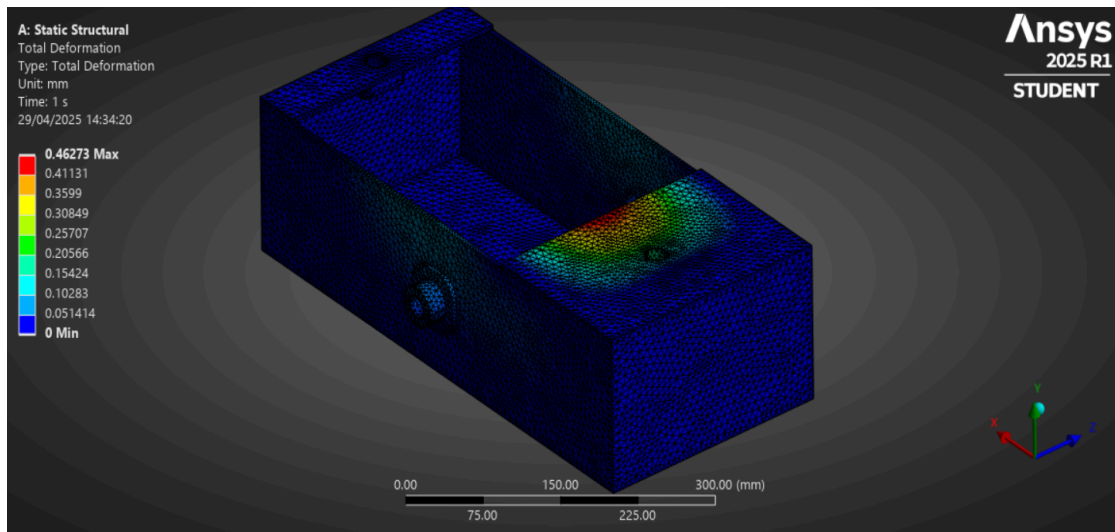


Figure 4.3: Total Deformation.

The colors represent the total displacement or deformation experienced by the model due to the applied loads. The color scale of the image indicates the range of deformation, from 0 mm (blue) to 0.46273 mm (red).

4.1.3.2 FOS

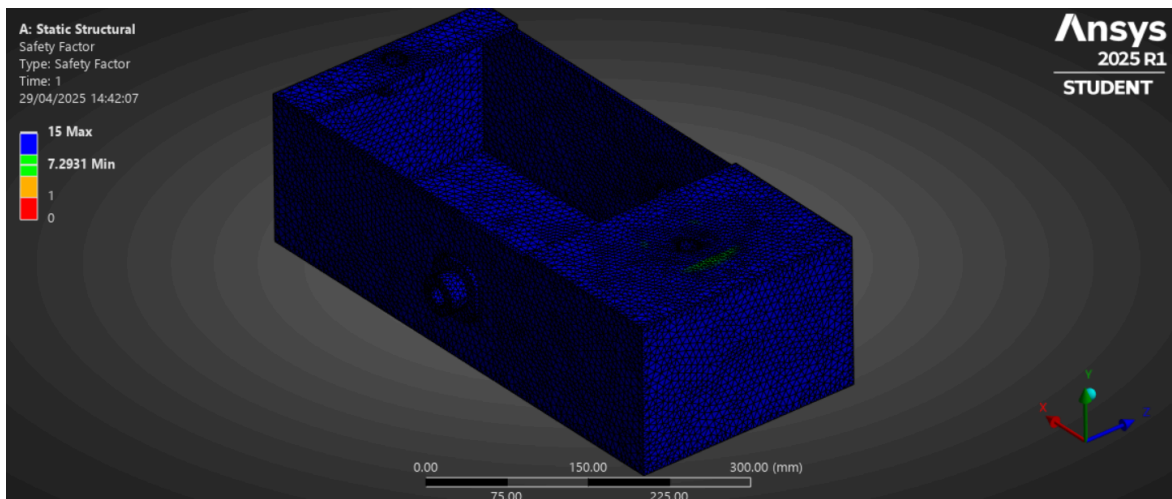


Figure 4.4: Factor of Safety.

The color map on the left represents the Safety Factor across the model. It ranges from a minimum value of 7.2931 (shown in green) to a maximum value of 15 (shown in blue).

A higher safety factor indicates a design that is less likely to fail under the applied loads, meaning the material can withstand more stress before yielding or breaking.

4.1.3.4 TOPOLOGY OPTIMIZATION OF UN-OPTIMISED DESIGN

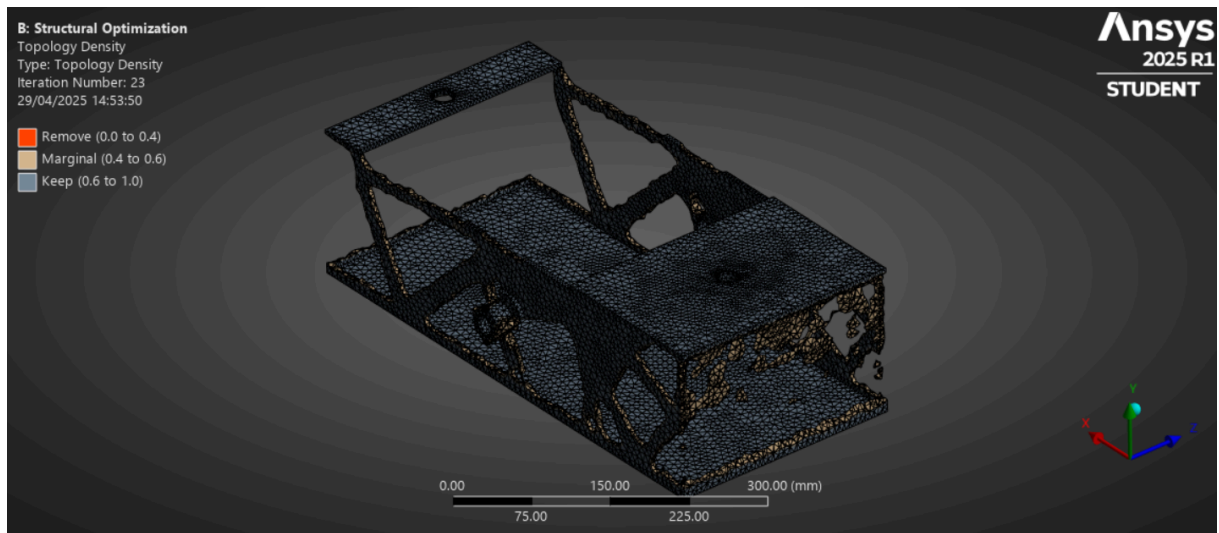


Figure 4.5: Topology Optimization.

Red Areas (0.0 to 0.4): These areas have low density, meaning material is removed from these regions in the optimization process. The red regions represent parts of the structure where material is deemed unnecessary for maintaining structural integrity.

Yellow Areas (0.4 to 0.6): These are **marginal regions** where some material is kept, but the density is relatively low. These areas may still carry some load, but further optimization could potentially reduce material further.

Grey (0.6 to 1.0): These areas have the highest density, meaning material is kept here to ensure the strength and stability of the structure. These regions are critical for maintaining the structural integrity under the applied loads.

The goal of this topology optimization is to minimize material usage while ensuring that the structure retains its mechanical performance. The optimization algorithm removes material from regions where it is not needed (shown in red), while keeping material where it is most needed (shown in grey).

This process is ideal for creating lightweight, yet strong, structures like chassis, as material is only kept where necessary, improving efficiency and reducing weight.

4.1.4 OPTIMISED DESIGN

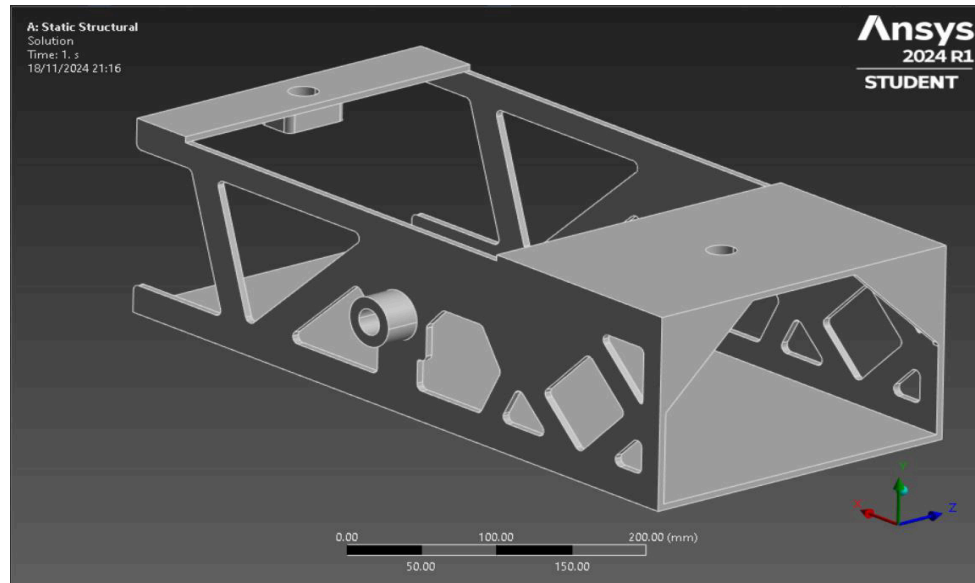


Figure 4.5: Optimised design.

The structure has undergone topology optimization, which has removed material from areas where it wasn't contributing significantly to the strength or stability of the design.

The optimized design features hollow regions and lightweight sections, ensuring that material is retained where it is most needed for load-bearing purposes, while non-critical regions are simplified or removed.

It is made using sheet metal which is lazer-cut and welded together.

NOTE:

The meshing and forces considered are similar as mentioned above.

4.1.5 ANALYSIS OF OPTIMISED DESIGN

4.1.5.1 TOTAL DEFORMATION

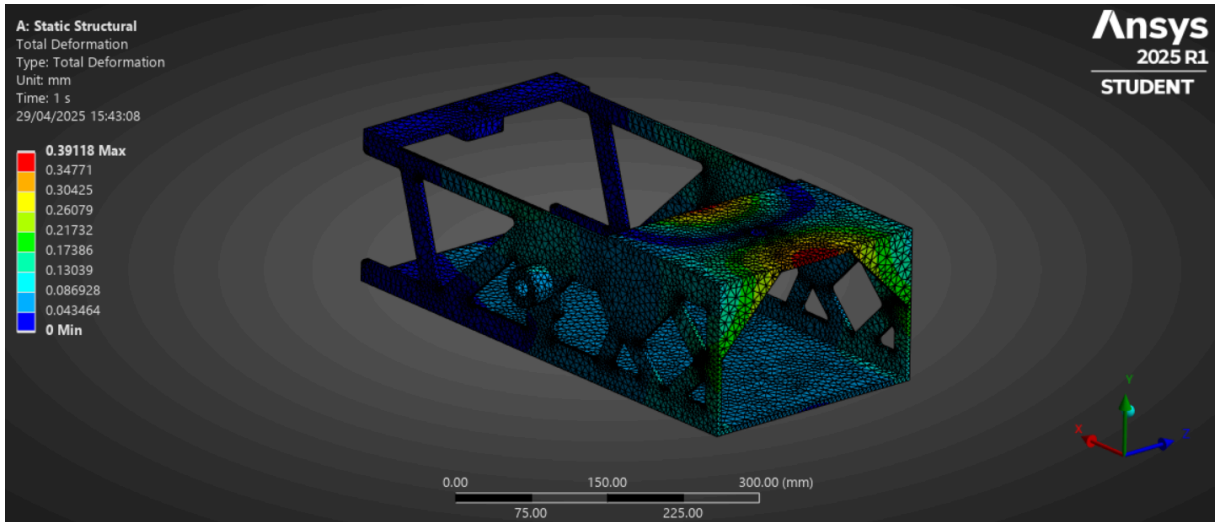


Figure 4.6: Deformation of optimised design.

Max Deformation: The highest deformation observed is **0.39118 mm** (shown in red), which indicates the area with the most movement or displacement due to the applied forces.

Min Deformation: The lowest deformation is **0 mm** (shown in blue), indicating areas that experience no deformation, either because they are fixed or under negligible load.

4.1.5.1 FOS

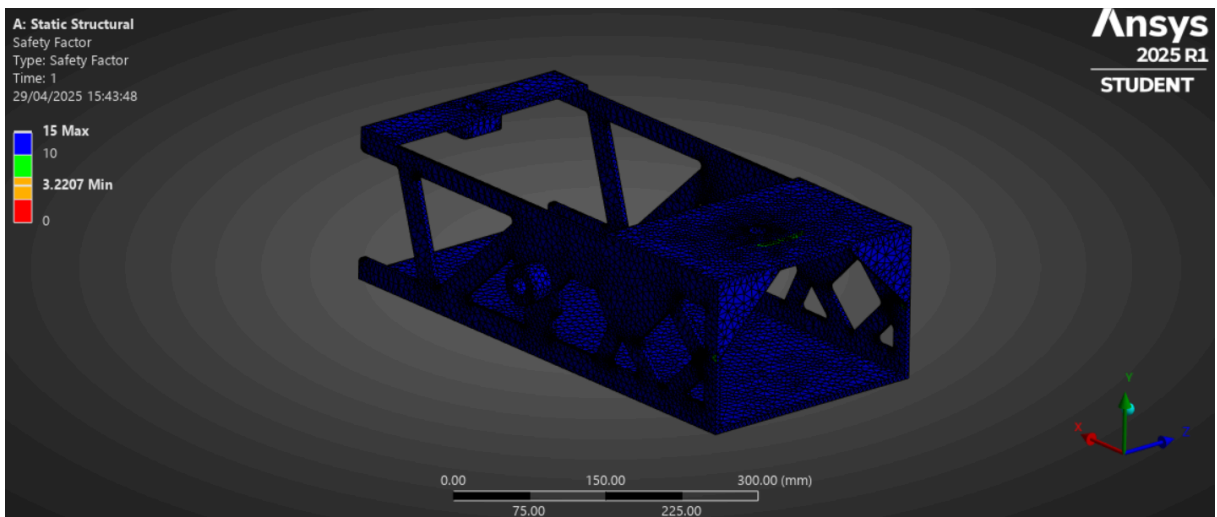


Figure 4.7: FOS of optimised design.

Max Safety Factor: The maximum safety factor is **15** (shown in blue), meaning the design is well within safe limits in these regions. A high safety factor implies that the material can withstand much more stress than it is currently subjected to, indicating a high margin of safety.

Min Safety Factor: The minimum safety factor is **3.2207** (shown in yellow). This is still considered safe since the factor is well above **1** (a safety factor of **1** would indicate the material is at its yield point). The structure is deemed safe in all regions.

4.2 DIFFERENTIAL

4.2.1 MESHING

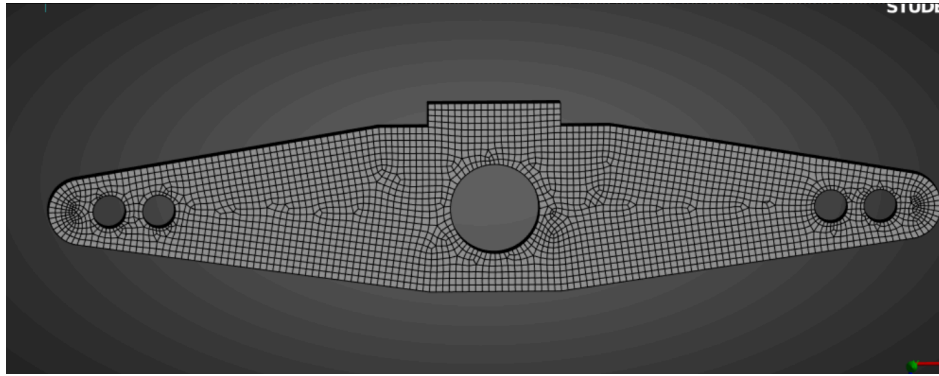


Figure 4.8 : Meshing on differential

The mesh for this model was created with a 2 mm element size, which allows for a finer resolution while ensuring that the component's behavior is accurately captured under applied loads. This finer discretization allows the software to numerically solve the physical equations governing the model's behavior, such as stress, strain, and deformation, under various applied loads. A finer mesh, while providing more accurate results, also requires more computational resources. The mesh allows for simulating and analyzing how the structure will respond to forces, moments, and boundary conditions, which is crucial for understanding the component's performance under real-world conditions.

In terms of mesh statistics, the model contains 27,000 elements, with each element being 2 mm in size. The number of elements defines the resolution of the mesh, where more elements lead to a more detailed and accurate simulation, but at the cost of increased computational demand.

4.2.2 FORCES ACTING

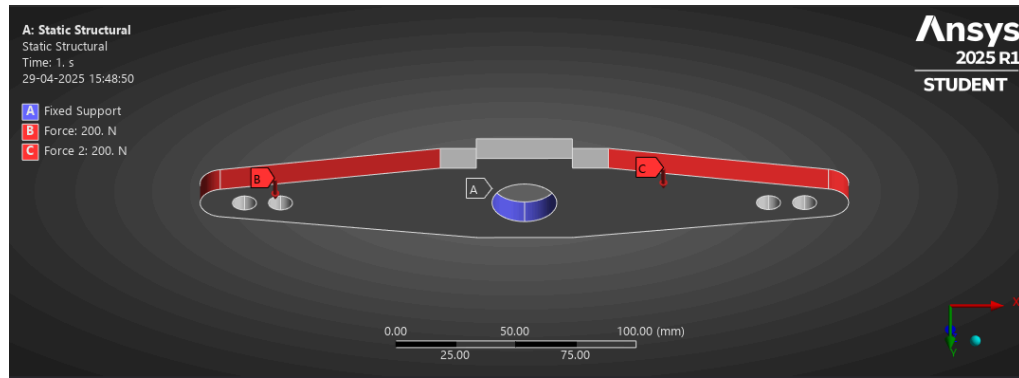


Figure 4.9 : Boundary conditions on differential

Figure 4.9 shows the boundary conditions applied to the model in ANSYS for a static structural analysis. The model is subjected to the following conditions:

1. **Fixed Support (A):** Located at one end of the component, it restricts all degrees of freedom (translation and rotation), preventing any movement at that location.
2. **Force 1 (B):** A **200 N force** is applied at a specified point (B) on the structure.
3. **Force 2 (C):** Another **200 N force** is applied at a second point (C), which is in the opposite direction or aligned similarly to the first force.

When a rover moves over rough terrain, the differential absorbs the forces transmitted from the suspension arms. These forces can be a combination of the weight of the rover (or parts of it), the reaction forces from the wheels, and the dynamic forces generated by obstacles.

4.2.3 ANALYSIS FOR UN-OPTIMISED DESIGN

4.2.3.1 TOTAL DEFORMATION

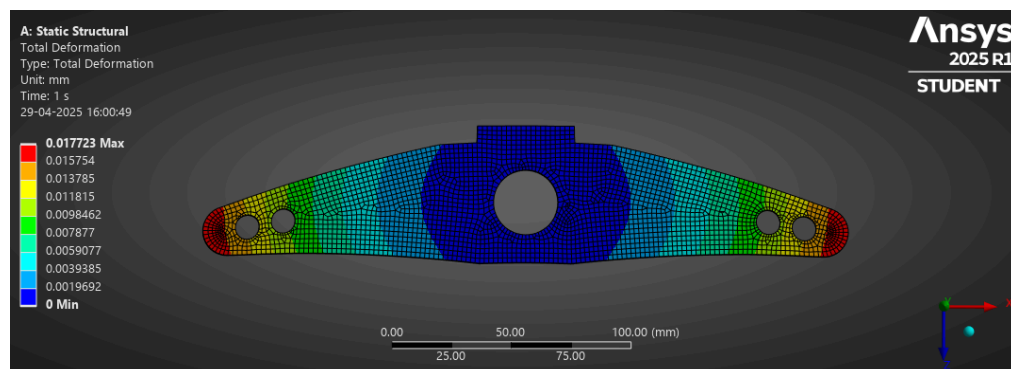


Figure 4.10 : Total Deformation

Figure 4.10 shows the total deformation of the differential under applied loads. The maximum deformation of 0.017723 mm occurs in the central region of the component, while the rest of

the model experiences minimal deformation. The color gradient indicates varying levels of displacement, with the red region showing the highest deformation and the blue region the lowest.

4.2.3.2 EQUIVALENT STRESS

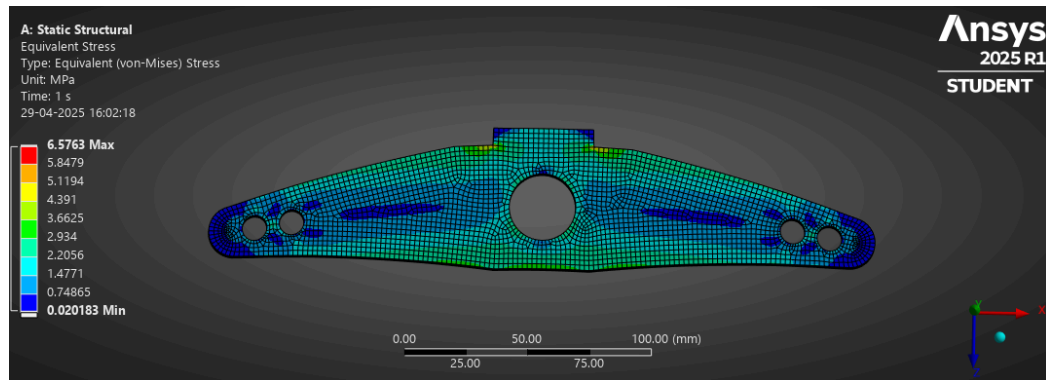


Figure 4.11 : Total Deformation

Figure 4.11 displays the equivalent von-Mises stress distribution across the differential. The maximum stress of **6.5763 MPa** is located near the central area where the forces are applied, while the rest of the component shows lower stress values.

4.2.3.3 TOPOLOGY OPTIMIZATION OF UN-OPTIMISED DESIGN

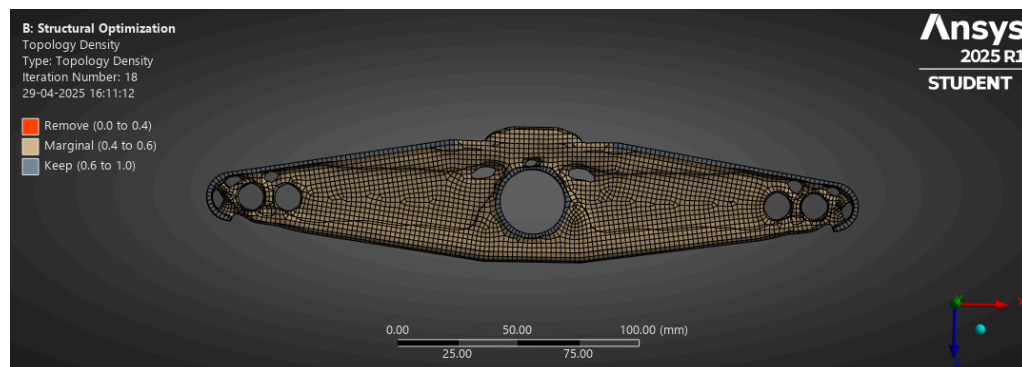


Figure 4.12 : Topology optimization of differential

Figure 4.12 shows the topology optimization of the differential component. The topology density map is displayed in three categories:

- Red (0.0 to 0.4): Regions that are considered unnecessary material, which will be removed during optimization.
- Yellow (0.4 to 0.6): Marginal areas where material could be reduced, but may be kept depending on performance requirements.
- Grey (0.6 to 1.0): Critical areas that must be kept for structural integrity and performance.

4.2.4 OPTIMISED DESIGN

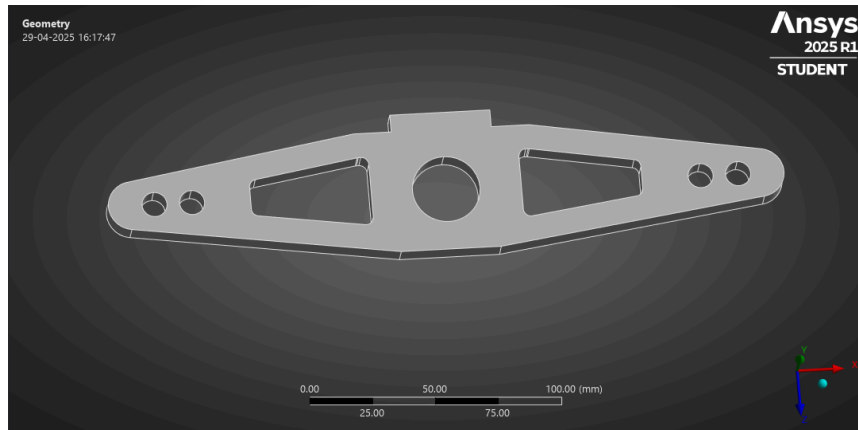


Figure 4.13 : Optimized design

Figure 4.13 shows the optimized geometry of the differential component after the topology optimization process. The material has been removed from the low-stress areas (as indicated by the triangular cutouts and reduced thickness) to minimize weight while retaining the essential structural integrity needed for performance. The result is a lighter and more efficient design, focused on maintaining strength where it's needed most, such as around the mounting holes and central support.

4.2.5 ANALYSIS OF OPTIMISED DESIGN

4.2.5.1 TOTAL DEFORMATION

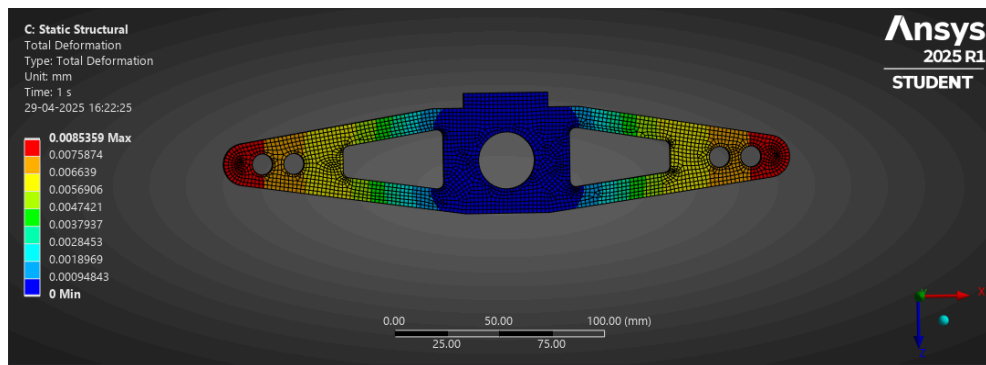


Figure 4.14 : Total deformation

Max deformation : 0.0085 mm

Min deformation : 0 mm

4.3 ROBOTIC ARM

4.3.1 MESHING

Meshing was done at an element size of 2 mm, with mixed elements, so as to capture the surfaces accordingly. A mesh convergence study was performed to ensure the solution does not change with mesh size.



Figure 4.15 : Meshing on Link

4.3.2 BOUNDARY CONDITIONS

Links -

Assuming the analysis for the shoulder joint, the lower bearing insert was treated as fixed support and the other insert was subjected to the torque mentioned in the above calculations.

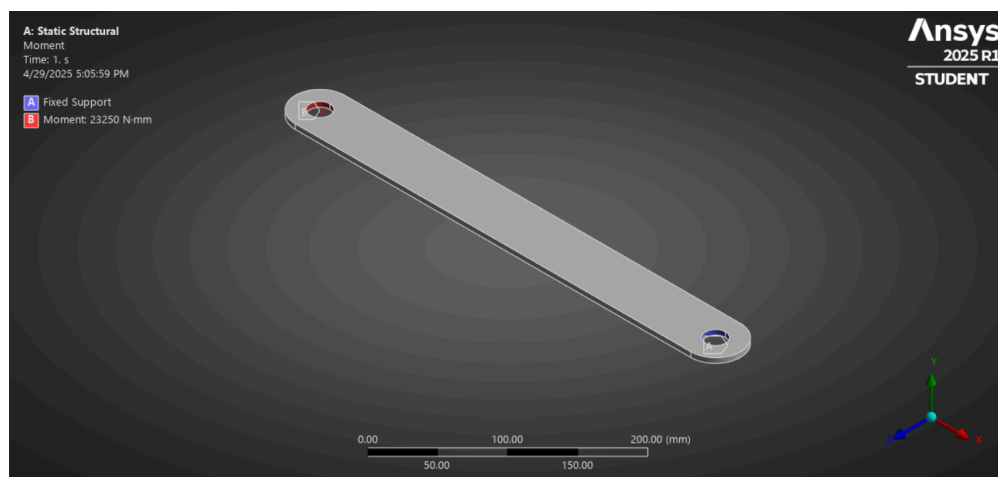


Figure 4.16 : Boundary conditions on links

4.3.3 RESULTS FOR UNOPTIMIZED STRUCTURE

4.3.3.1 TOTAL DEFORMATION

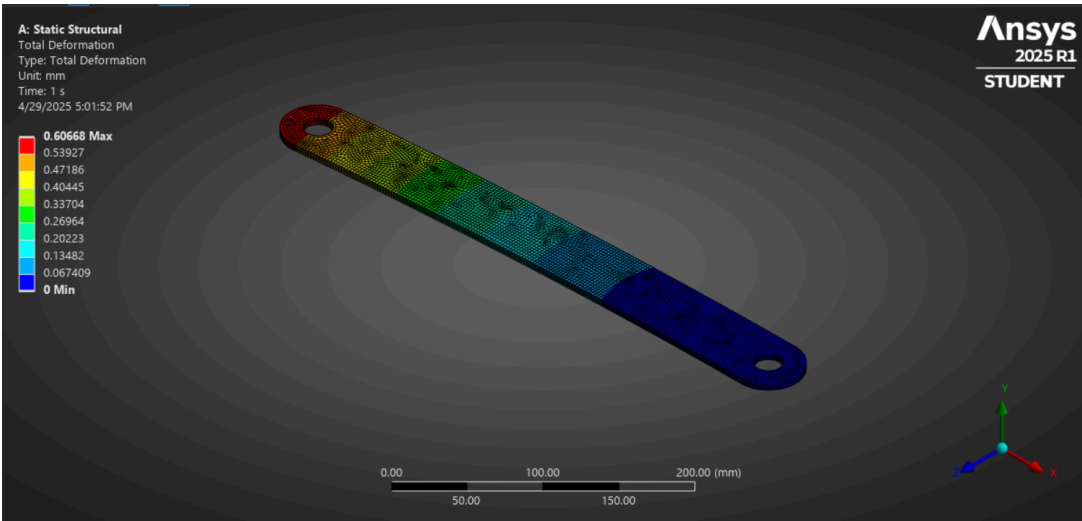


Figure 4.17 : Total deformation of links

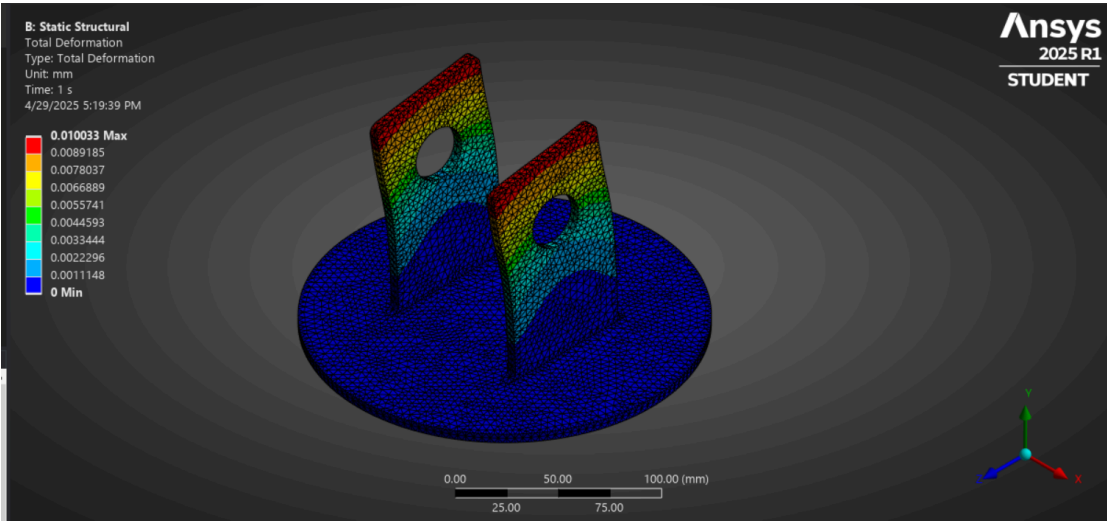


Figure 4.18 : Total deformation of base

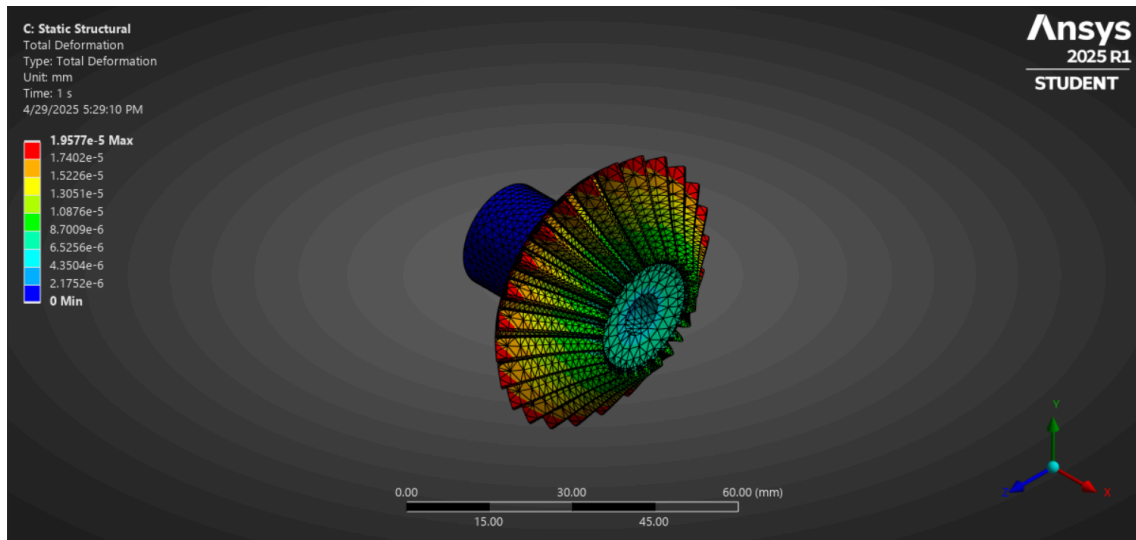


Figure 4.19 : Total deformation of gear

4.3.3.2 MAX EQUIVALENT STRESS

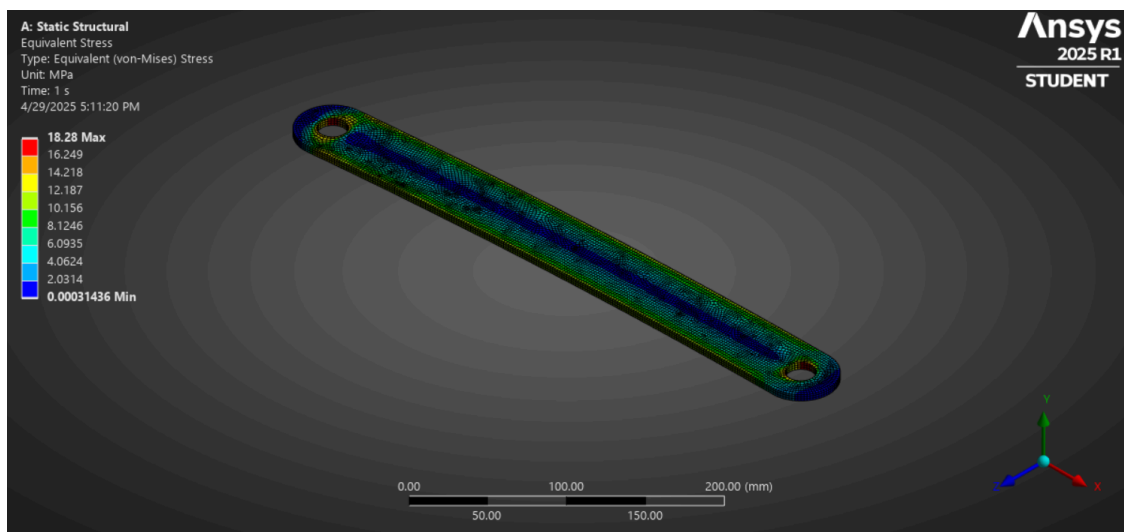


Figure 4.20 : Equivalent stress on links

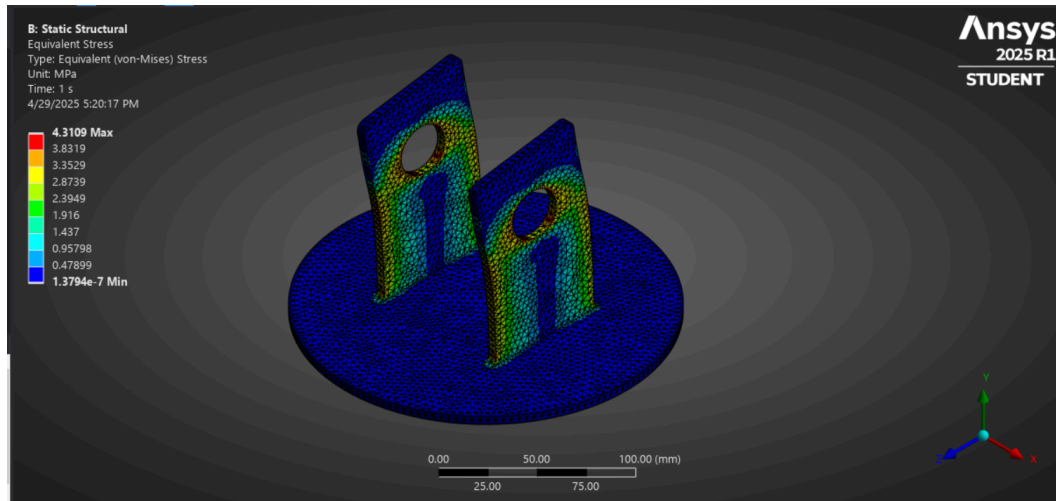


Figure 4.21 : Equivalent stress on base

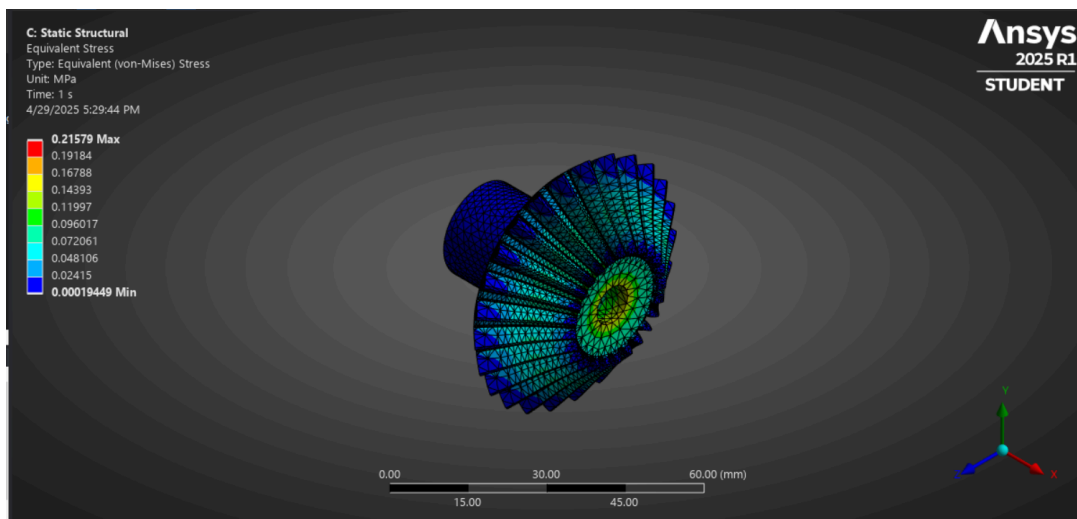


Figure 4.22 : Equivalent stress on gear

Thus, the overall structure was safe as per the initial material selection based analysis. Factor of safety obtained was 15 for all components. The further study would be upon improving the material and topology, so as to decrease the weight, while maintaining the same performance.

4.3.4 TOPOLOGY OPTIMIZATION

ANSYS Workbench was used for reduction of weight of the individual component. Topology optimization was studied for links and base structure. Target was set for weight reduction by 60 % for individual components.

Based on the applied load -

Link -

Initial Weight = 290.052 g

Final Weight = 178.273 g

% weight reduction = 38.4 %



Figure 4.23 : Topology Optimization

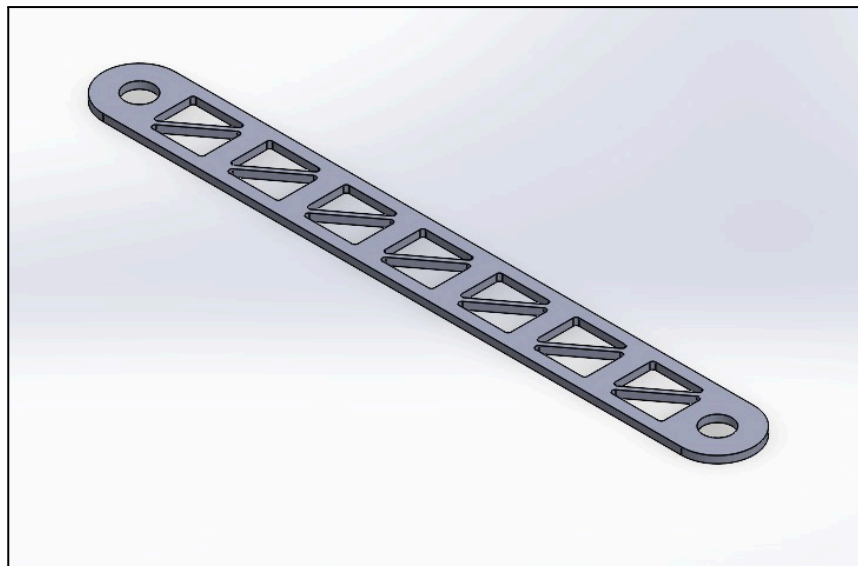


Figure 4.24 : Optimized design of link

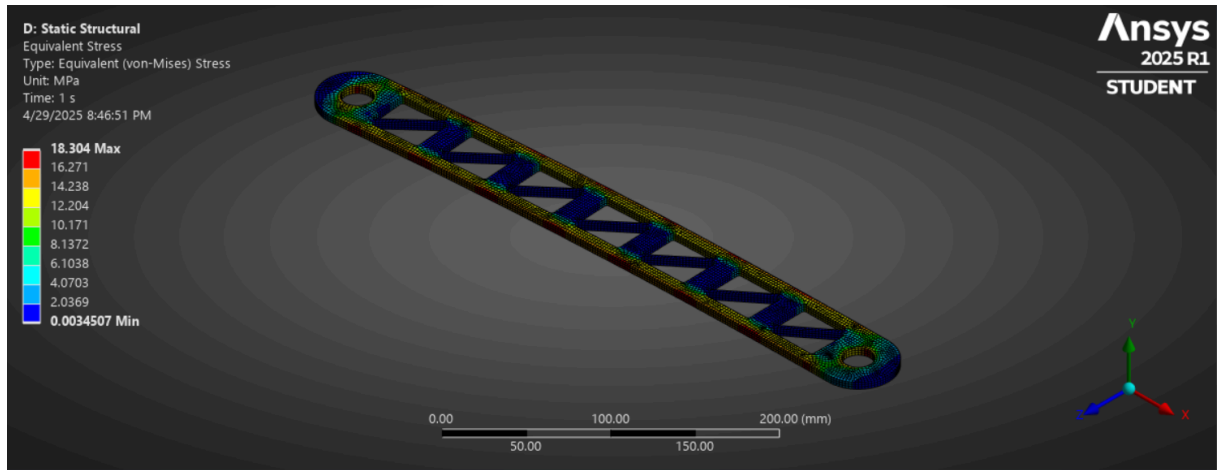


Figure 4.25 : Stress analysis on optimized design of link

Total Deformation	0.295 mm
Max. Von Mises Stress	18.34 MPa
FOS	13.658

Table 4.1 : Results (links)

Base -

Initial Weight = 653.124 g

Final Weight = 509.6 g

% weight reduction = 21.8 %

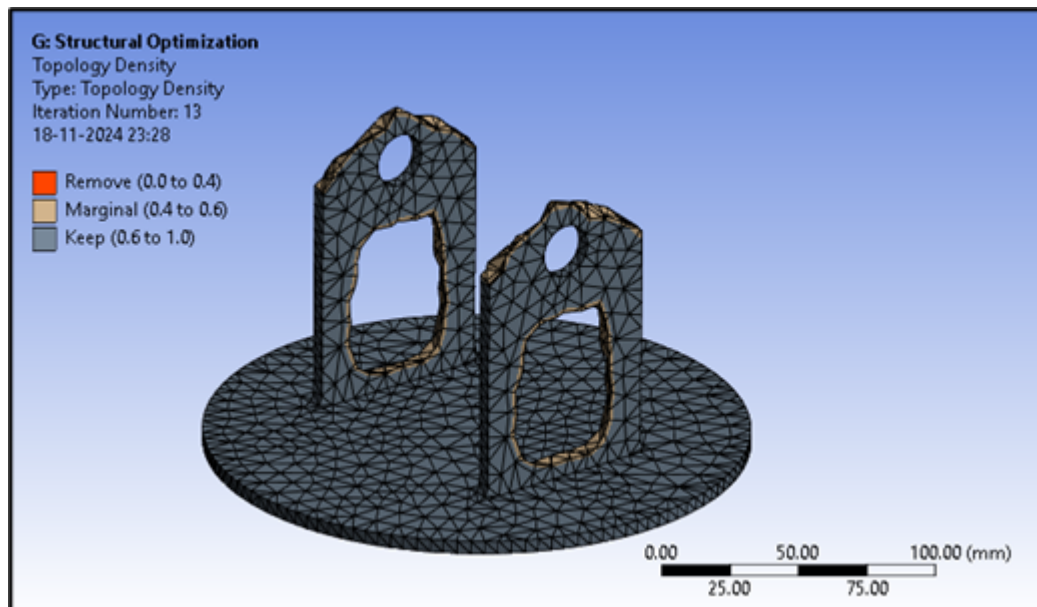


Figure 4.26 : Optimization of base

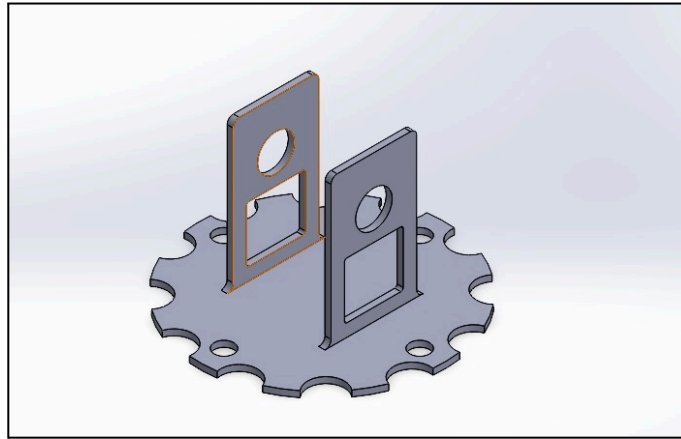


Figure 4.27 : Optimized design of base

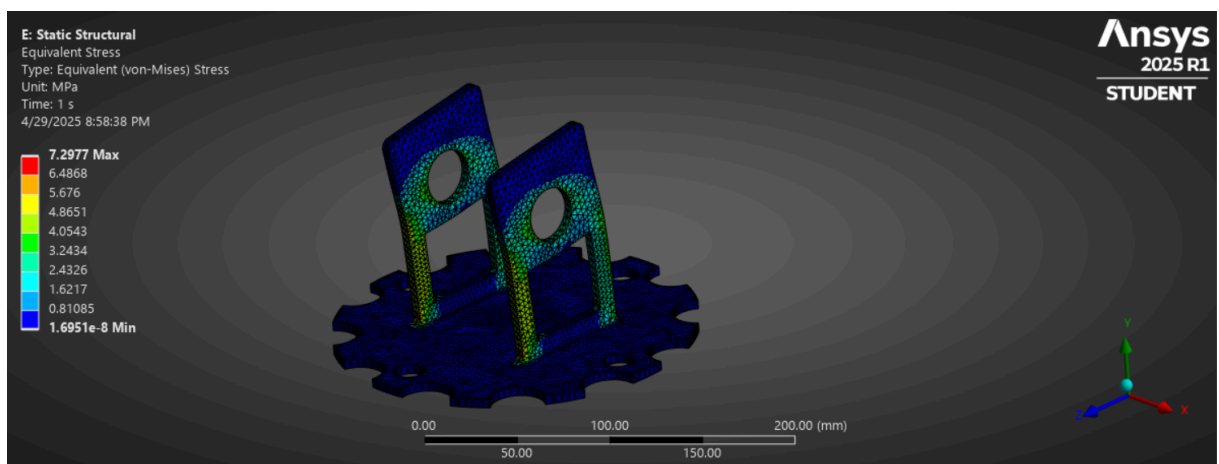


Figure 4.28 : Stress analysis on optimized design of base

Total Deformation	0.00453 mm
Max. Von Mises Stress	7.3 MPa
FOS	15

Table 4.2 : Results (base)

4.3.5 MATERIAL OPTIMIZATION

4.3.5.1 USE OF ADDITIVE MANUFACTURING MATERIALS FOR BEVEL GEARS

Initial iteration of the bevel gears were manufactured from stainless steel. Although this contributed to strength structurally, the major tradeoff was weight concentration at the end effector attachment point, which could lead to eventual failure of the holding structure.

After careful analysis of the scenario, bevel gears were concluded to rotate only for lightweight maintenance tasks and not for heavy payload based tasks. This allowed us to use lighter 3D printed bevel gears using lightweight, strong plastics.

Properties of ABS plastic -

Density = 1030 kg./m^3

$E = 2.09 \times 10^9 \text{ N/m}^2$

Poisson's ratio = 0.4089

Tensile Yield Strength = $2.744 \times 10^7 \text{ N/m}^2$

Tensile Ultimate Strength = $3.626 \times 10^7 \text{ N/m}^2$

Among many plastics, ABS was widely used as a material for such application because of its high strength and chemical resistance. Thus, the structural analysis was performed for ABS.

Results -

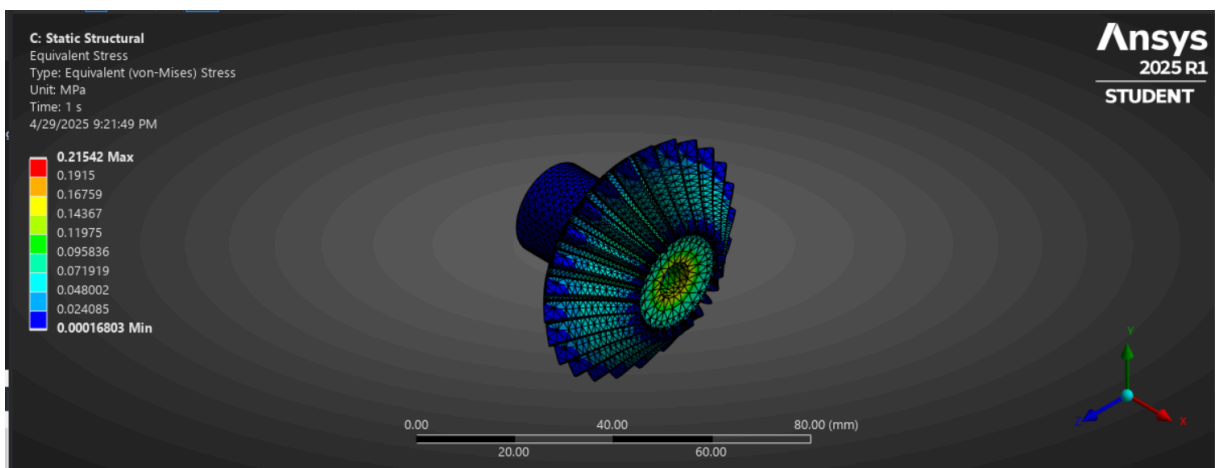


Figure 4.29 : Stress analysis on gear

Total Deformation	0.001763
Max. Von Mises Stress	0.21542 MPa
FOS	15

Table 4.3 : Results (gear)

Initial Weight = 180 g

Final Weight = 24 g

Thus, a weight reduction of 86.7 % was achieved by use of ABS as compared to steel.

4.3.5.2 COMPOSITE MATERIALS^[4]

The use of composites for student competitions has increased by a significant percentage over the years. Composites allow lightweighting of structures without topology optimization and also have automated methods of custom manufacturing. The only tradeoff is the cost of employing the materials into the system. However, a case study for the robotic arm links is simulated to show the effectiveness of composites and also to study the characteristics of the composites structure under loading.

Theory -

A composite or composite material (also composition material) is a material which is produced from two or more constituent materials. These constituent materials have notably dissimilar chemical or physical properties and are merged to create a material with properties unlike the individual elements. Within the finished structure, the individual elements remain separate and distinct, distinguishing composites from mixtures and solid solutions. Composite materials with more than one distinct layer are called composite laminates.

Typical engineered composite materials are made up of a binding agent forming the matrix and a filler material (particulates or fibres) giving substance, e.g.:

- Concrete, reinforced concrete and masonry with cement, lime or mortar (which is itself a composite material) as a binder
- Composite wood such as glulam and plywood with wood glue as a binder
- Reinforced plastics, such as fiberglass and fibre-reinforced polymer with resin or thermoplastics as a binder
- Ceramic matrix composites (composite ceramic and metal matrices)
- Metal matrix composites

- advanced composite materials, often first developed for spacecraft and aircraft applications.

Composite materials can be less expensive, lighter, stronger or more durable than common materials. Some are inspired by biological structures found in plants and animals. Robotic materials are composites that include sensing, actuation, computation, and communication components.

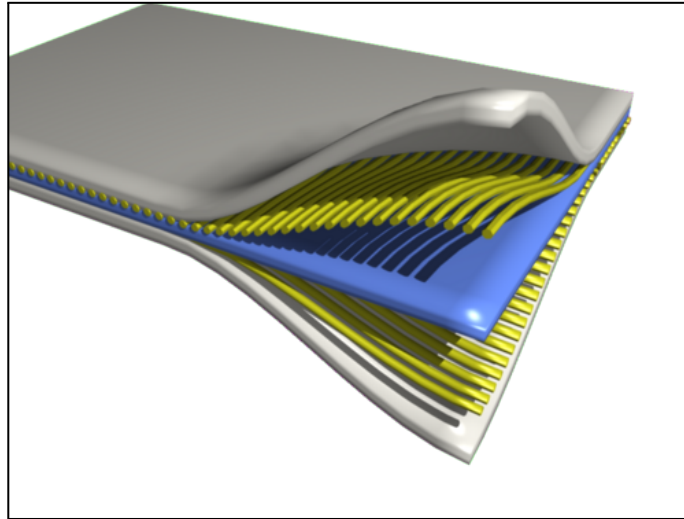


Figure 4.28 : Composite Materials

Simulation Study -

A simulation is performed for a single link in ANSYS ACP Composite Analysis Software. Two orientations - longitudinal and cross ply are studied and the conclusions are made on the basis of the stress analysis.

Carbon Fiber (1.5 mm)
Aluminium (1.5 mm)
Carbon Fiber (1.5 mm)

Basic structure of the composite

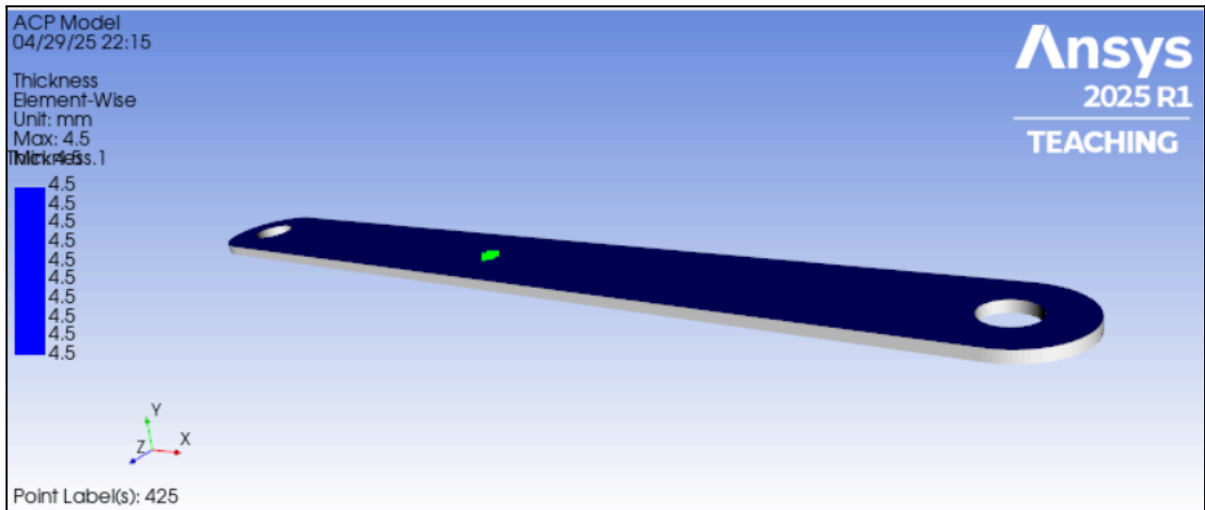


Figure 4.29 : ACP analysis of composite

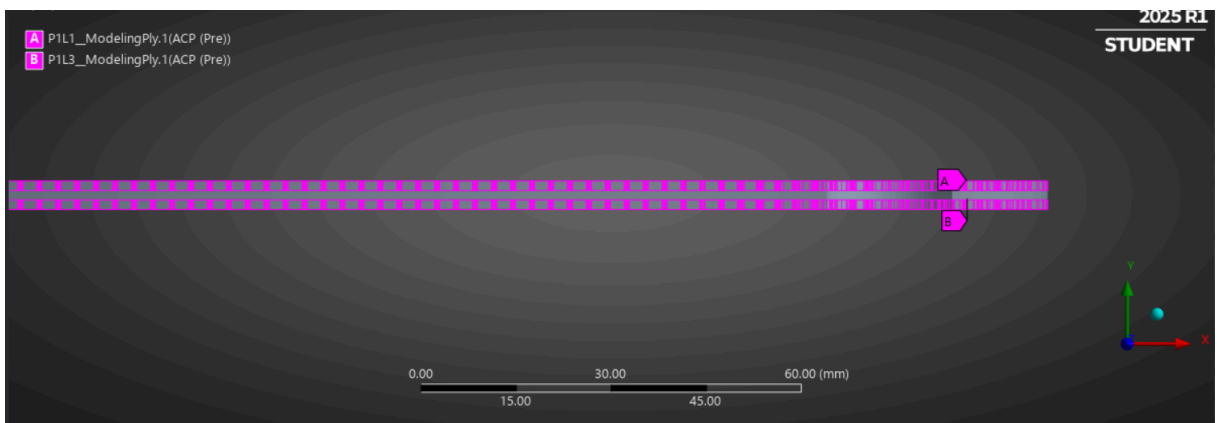


Figure 4.30 : Ansys Analysis

Analysis -

1) Longitudinal Ply

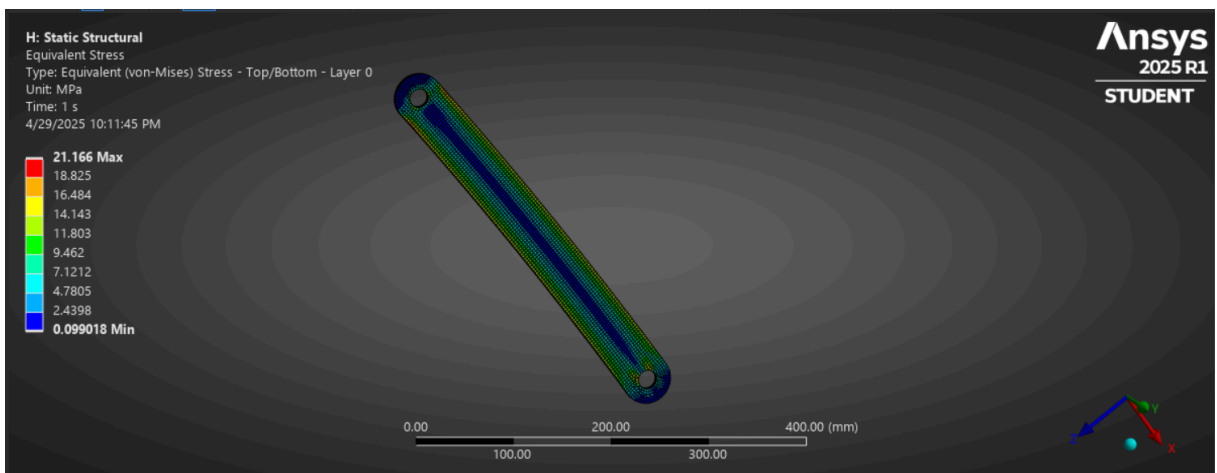


Figure 4.31 : Stress Analysis (Longitudinal Ply)

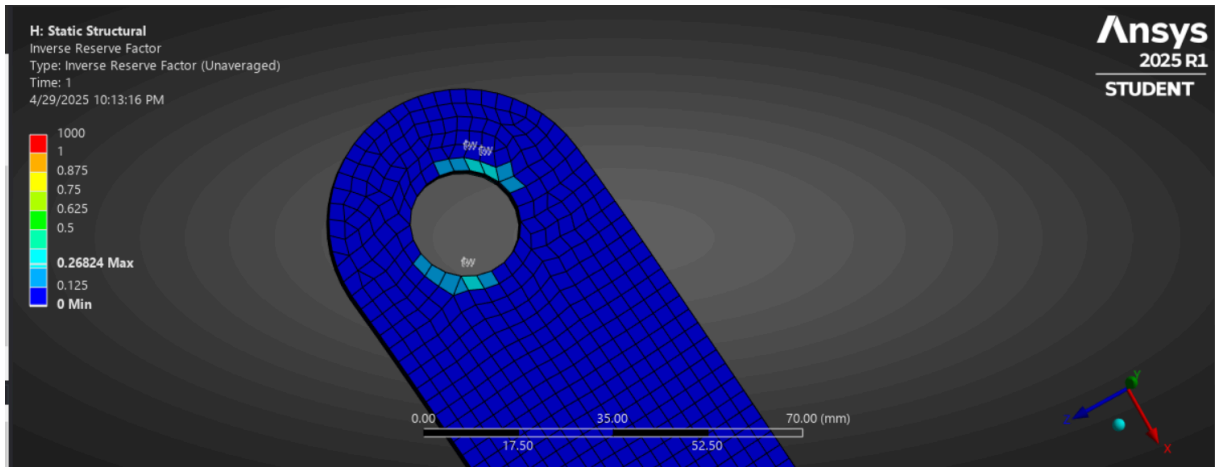


Figure 4.32 : Inverse reserve factor

Total Deformation	0.514 mm
Max. Von Mises Stress	21.17 MPa
Inverse Reserve Factor	0.268

Table 4.4 : Results (Longitudinal Ply)

The term ‘tw’ indicates wrinkling of the top face sheet.

2) Cross Ply

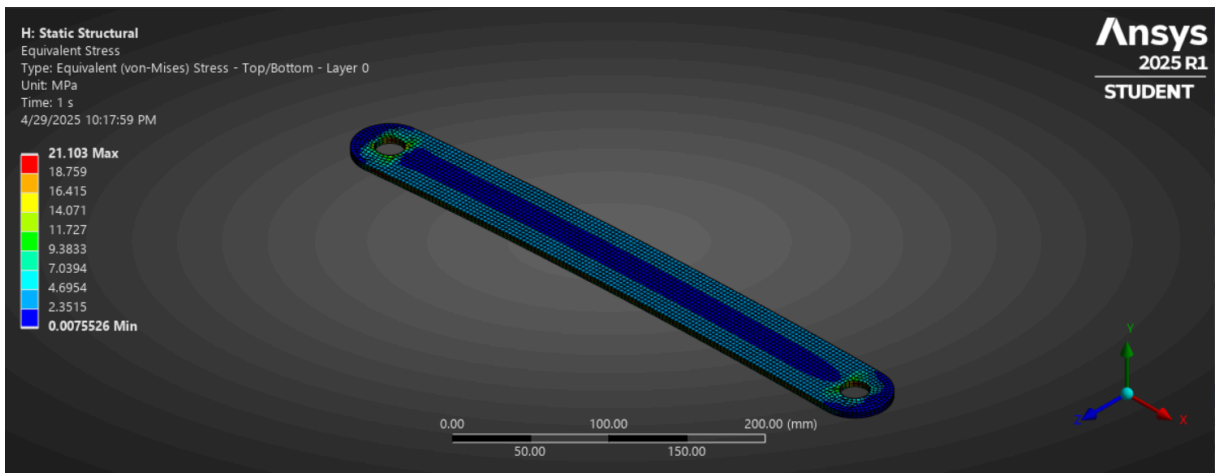


Figure 4.33 : Stress Analysis (Cross Ply)

Total Deformation	1.5116 mm
Max. Von Mises Stress	21.103 MPa
Inverse Reserve Factor	0.1752

Table 4.5 : Results (Cross Ply)

Conclusion -

Inverse Reserve Factor result type is an inverse margin to the Safety Factor. The failure load can be defined as the load value divided by IRF. Failure is experienced when you have an Inverse Reserve Factor greater than 1.

$$\text{IRF} = 1/\text{SF}$$

The critical values of reserve factors lie between zero and one, whereas the non-critical values range from one to infinity. Whether the results are shown in numeric form or as contour plots, the non-critical values tend to be emphasized in comparison to critical values.

For the longitudinal ply case, the stress obtained is higher and deformation is lower, due to plies acting in the same direction. However, in cross ply case, the stress remains almost the same, due to cross-distribution of load, thus reducing the possibility of wrinkling in the top face. Deformation in longitudinal direction is higher due to cross-ply structure.

4.4 SUSPENSION

Based on the same mesh size as used for the robotic arm link, the unoptimized structure was analyzed initially, for the given load calculations.

Results -

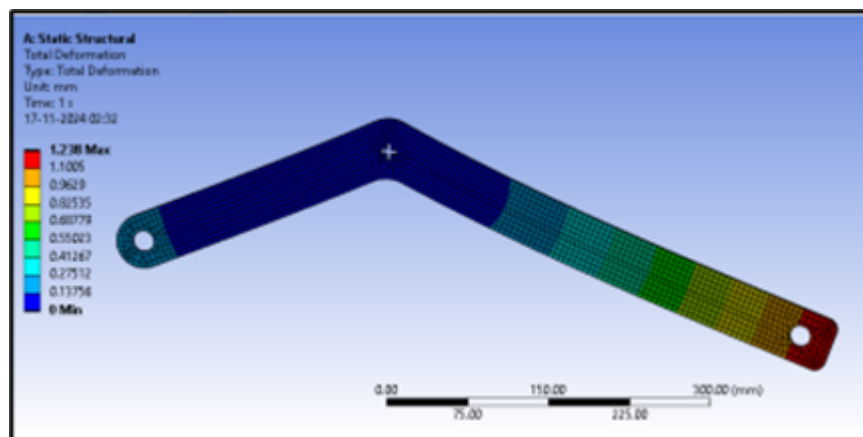


Figure 4.34 : Total deformation for rocker

Deformation (mm)	1.238
Stress (MPa)	40.016
FOS	6.9971

Table 4.6 : Results (Suspension Links)

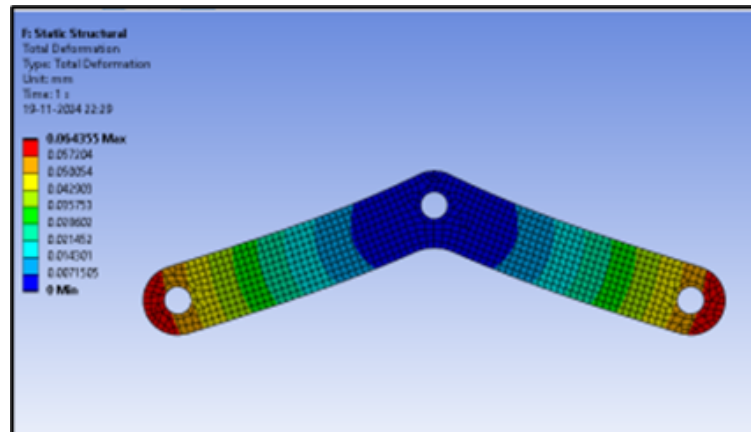


Figure 4 35. : Total deformation for bogie

Deformation (mm)	0.0644
Stress (MPa)	9.4141
FOS	15

Table 4.7 : Results (Bogie)

Optimization -

The same procedure of optimization was followed for suspension links, and were analyzed in ANSYS Workbench, against the same parameters for the unoptimized structure.

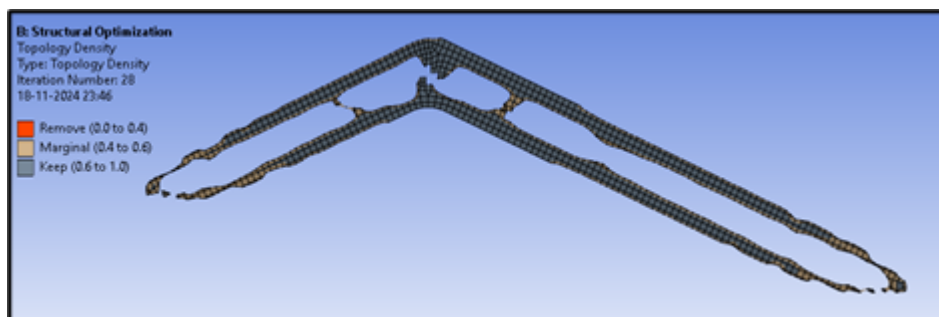


Figure 4.36 : Topology optimization on rocker

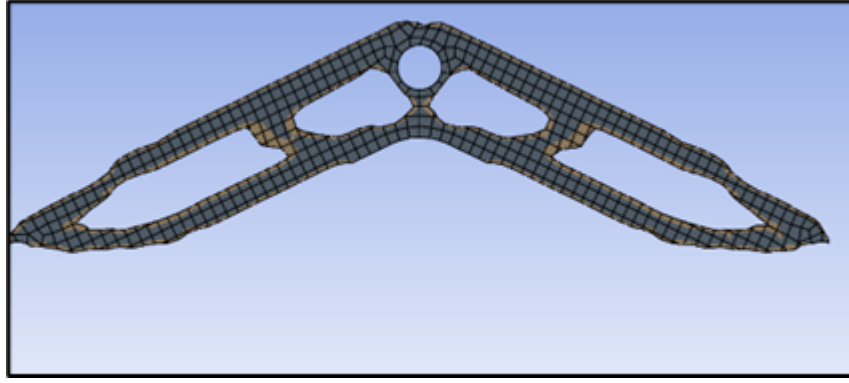


Figure 4.37 : Topology optimization on bogie

Results for optimized structure -

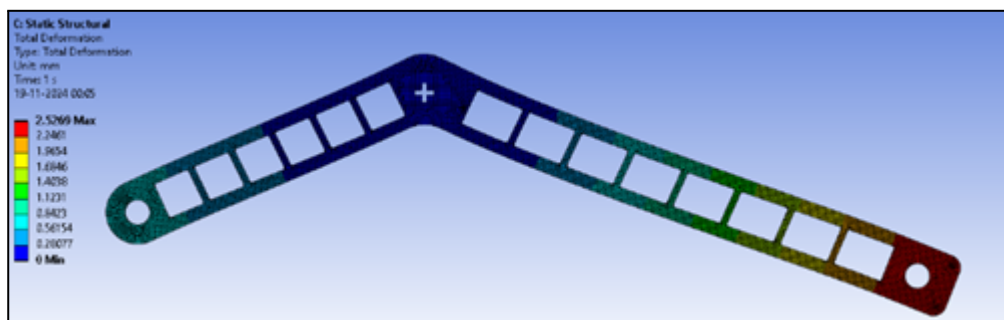


Figure 4.38 : Total deformation for optimized suspension link

Deformation (mm)	2.527
Stress (MPa)	135.12
FOS	2.072

Table 4.8 : Results (Optimized Suspension link)

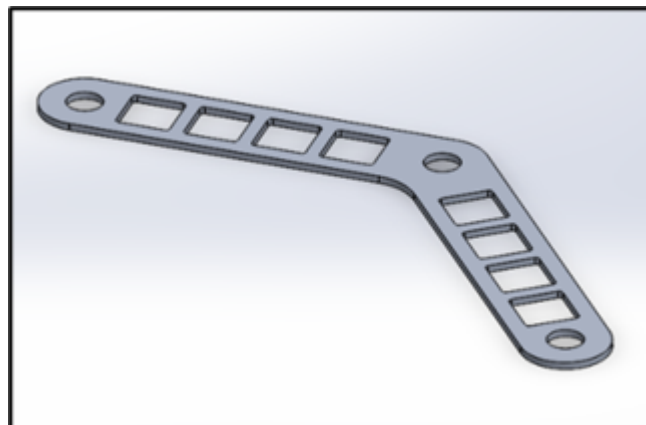


Figure 4.39 : Optimized design

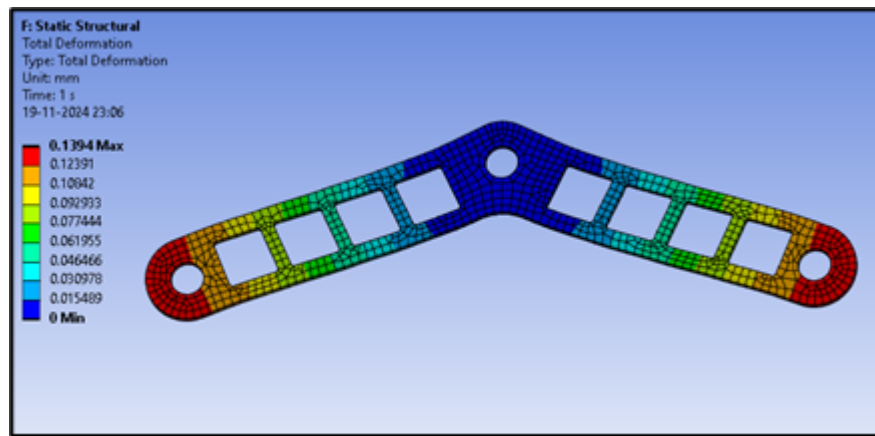


Figure 4.40 : Total deformation for optimized bogie

Deformation (mm)	0.1394
Stress (MPa)	23.468
FOS	11.931

Table 4.9 : Results (Optimized bogie)

CHAPTER 5: CONCLUSION

5.1 CHASSIS

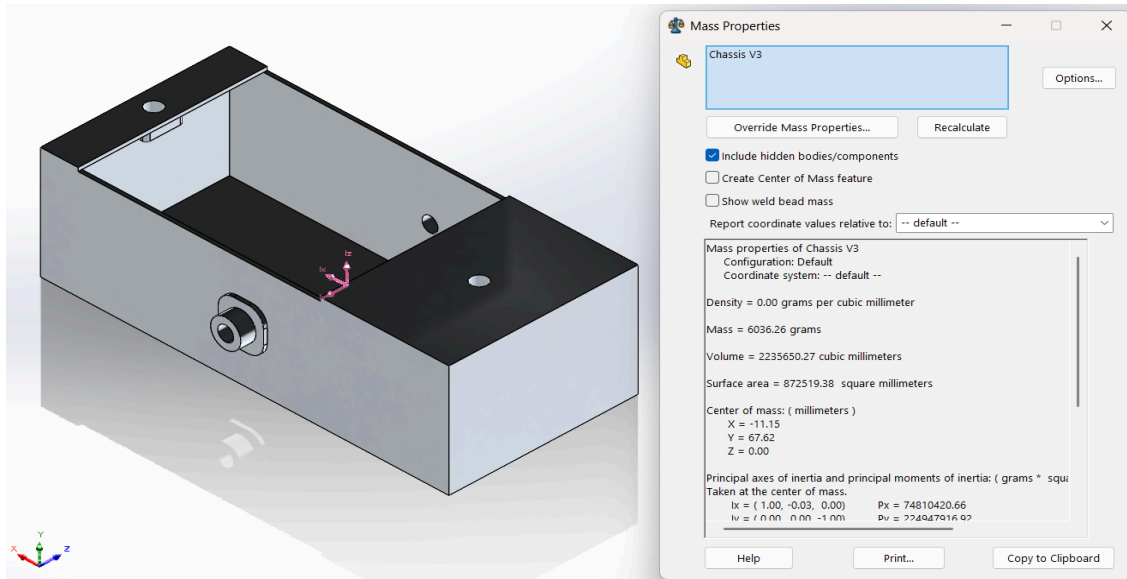


Figure 5.1: Weight of original design (6.036 Kg).

The optimized rover chassis design demonstrates significant improvements in weight reduction and performance metrics, reflecting the success of the topology optimization process. The weight of the chassis decreased from 6.036 kg to 4.2 kg, achieving a remarkable 30% reduction in weight. This aligns well with the design objective of reducing material usage while maintaining structural integrity.

The factor of safety (FoS) decreased from 7.29 to 3.22, which is still within the acceptable range of 3 to 4, as per the design requirements. This reduction in the FoS indicates that the material usage is now more efficient, with less redundant material contributing to the structure while still maintaining sufficient safety margins under operational loads.

The maximum deformation of the chassis improved slightly, decreasing from 0.46 mm to 0.39 mm.

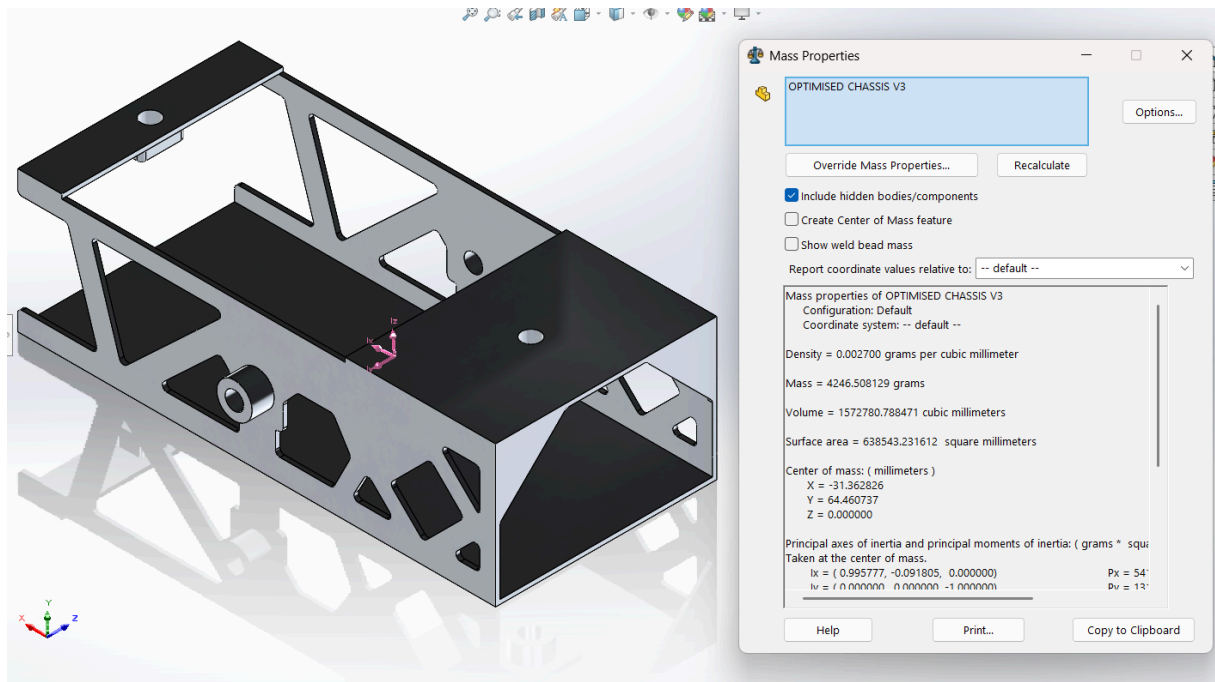


Figure 5.2: Weight of optimised design (4.2 Kg).

This reduction in deformation highlights that the optimized structure exhibits better stiffness and load-handling capacity despite the significant reduction in weight. This ensures that the chassis can endure operational forces without compromising its dimensional stability.

In summary, the optimization process achieved a lightweight yet robust design, meeting the critical performance objectives of weight reduction, safety, and stiffness. These improvements make the chassis highly suitable for the rover's demanding operational and competitive environments.

5.2 DIFFERENTIAL

The optimization process for the differential resulted in significant improvements in both deformation and mass. In the unoptimized state, the differential exhibited a maximum deformation of 0.017723 mm due to applied forces, particularly at the center of the component. Additionally, the unoptimized mass of the differential was 251.25 grams, with a volume of 93053.98 cubic millimeters and a surface area of 26529.89 square millimeters.

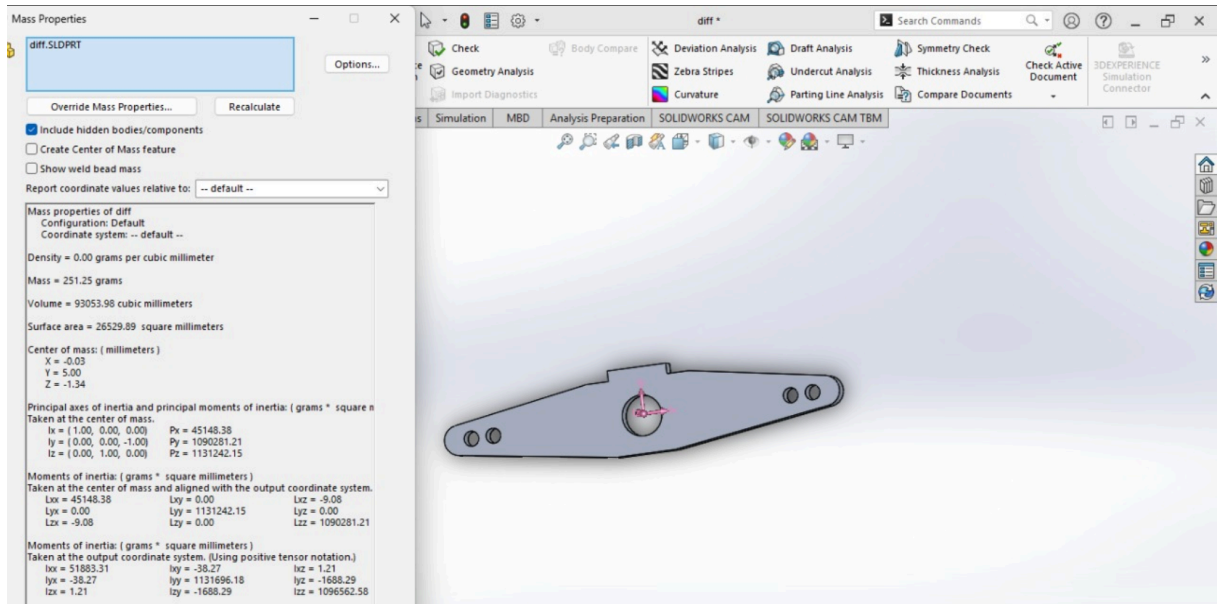


Figure 5.3: Weight of Original design

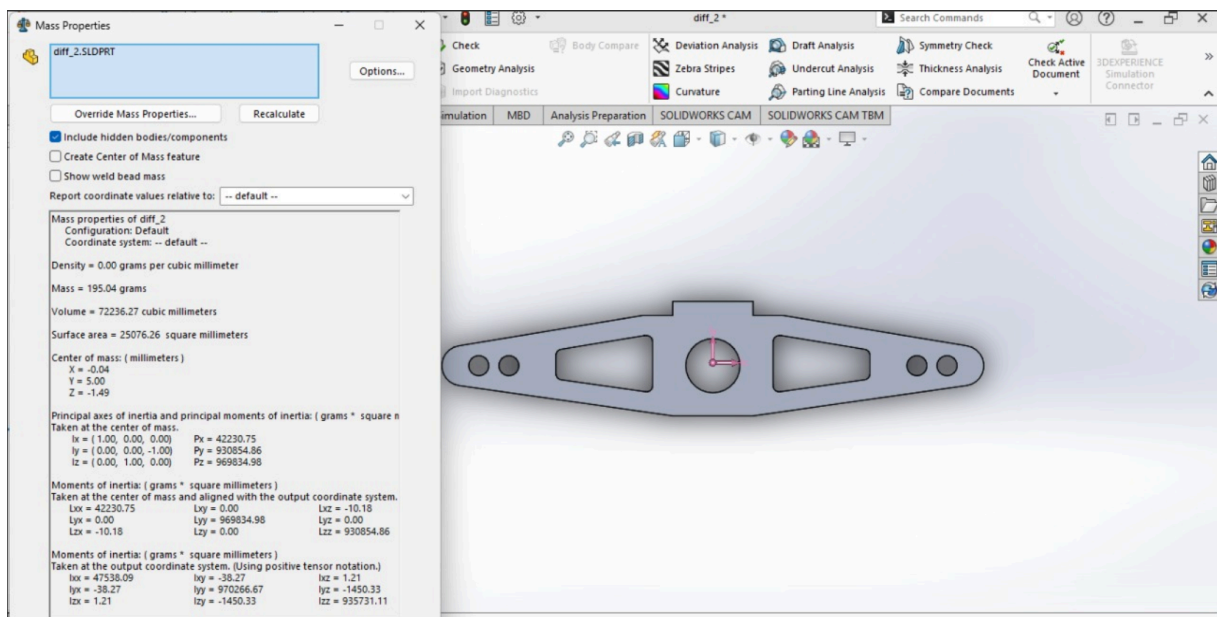


Figure 5.4: Weight of optimised design

Following the topology optimization process, the maximum deformation was reduced to 0.006859 mm, significantly improving the component's structural performance and reducing unnecessary movement under load. The optimization also resulted in a weight reduction, with the mass decreasing to 195.04 grams and the volume reducing to 72236.27 cubic millimeters, reflecting a more efficient use of material.

The reduction in weight contributes to the rover's overall mobility and energy efficiency, while the reduced deformation improves the differential's ability to handle the dynamic forces it encounters

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- [1] Rulebook-IRC-2023/2024
- [2] <https://beatty-robotics.com/differential-for-mars-rover/>
- [3] https://en.wikipedia.org/wiki/Robotic_arm
- [4] https://en.wikipedia.org/wiki/Composite_material